MIL-M-28787/112C(EC)
29 April 1983
SUPERSEDING
MIL-M-28787/112B(EC)
28 January 1980

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, SERIAL AND PARALLEL, DIGITAL, KEY CODE PDL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic serial and parallel shift register module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.215 failures per million operating hours for class I or 0.684 failures per million operatifng hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform tof the electrical requirements of class I only; and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PDL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/112-1 for class I and M28787/112-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

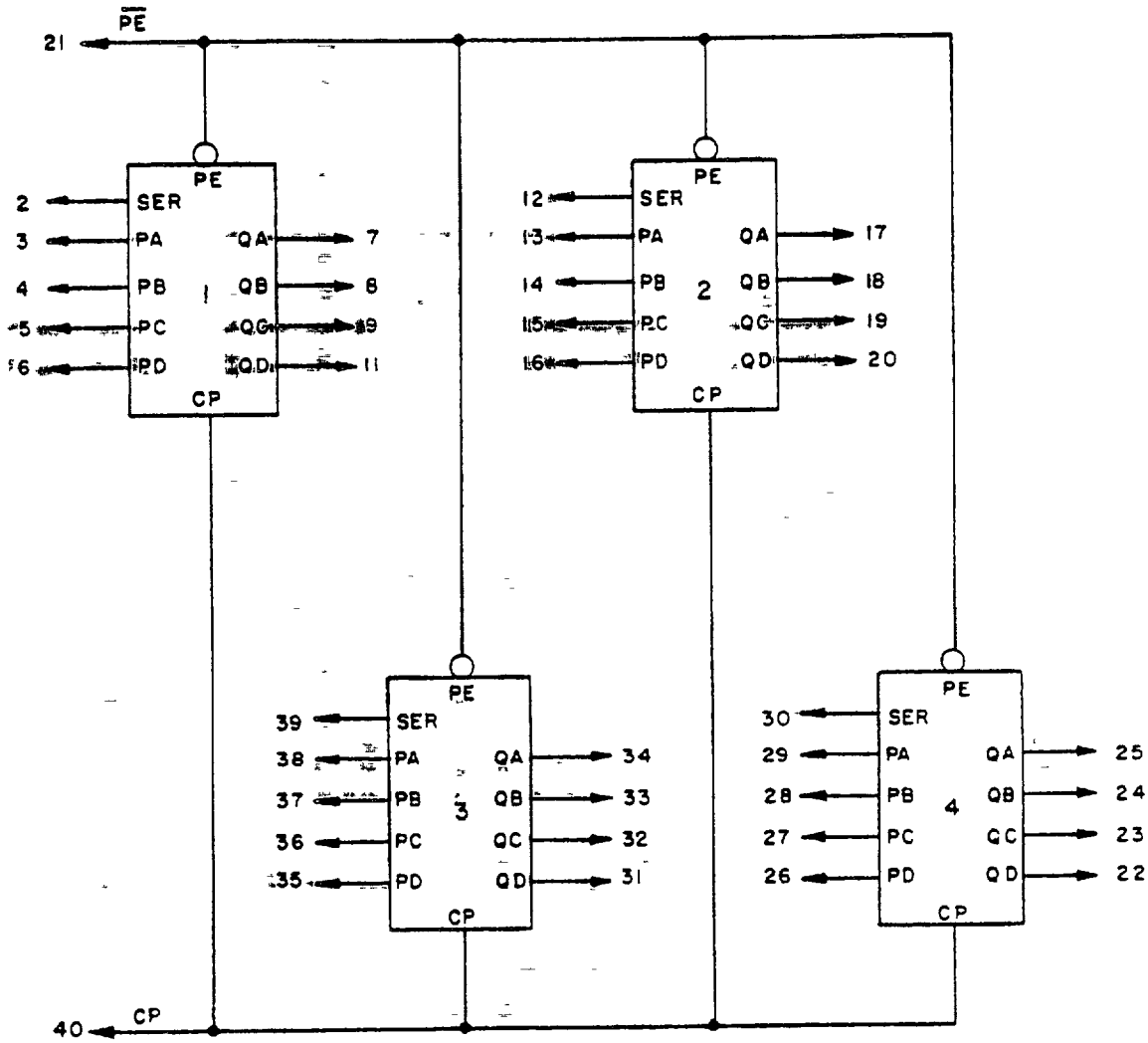
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTES:
1. $\overline{PE}$ = Common parallel enable.
2. CP = Common clock.
3. SER = Serial input
4. PA, PB, PC, PD = Parallel inputs
5. QA, QB, QC, QD = Parallel outputs.

* FIGURE 1. PDL module function diagram, serial and parallel shift register.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	V_{OUT} (low-loaded)	III	S	10	800	10	800	10	800	10	800	10	800	mV
B	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C4	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
C5-C20	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
C21,C22	Input breakdown current	V	S	0	4	0	4	0	4	0	4	0	4	mA
D1-D22	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E4	Input current, high level	VII	X	0	80	0	80	0	80	0	80	0	80	µA
E5-E20	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
E21	Input current, high level	VII	X	0	320	0	320	0	320	0	320	0	320	µA
E22	Input current, high level	VII	X	0	368	0	368	0	368	0	368	0	368	µA
F1-F4	Input current, low level	VIII	X	-0.10	-3.46	-0.10	-3.46	-0.10	-3.46	-0.10	-3.46	-0.10	-3.46	mA
F5-F20	Input current, low level	VIII	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
F21	Input current, low level	VIII	X	-1.0	-13.8	-1.0	-13.8	-1.0	-13.8	-1.0	-13.8	-1.0	-13.8	mA
F22	Input current, low level	VIII	X	-1.0	-15.9	-1.0	-15.9	-1.0	-15.9	-1.0	-15.9	-1.0	-15.9	mA
G1-G16	Output short circuit current	IX	X	-10	-80	-10	-80	-10	-80	-10	-80	-10	-80	mA
H1-H8	Functional shift right, low level and high level	X	X	◄----------				See table X					----------►	

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J20	Output voltage (input threshold conditions), low level	XI	X	0	400	0	400	0	400	0	400	0	400	mV
K1-K20	Output voltage (input threshold conditions), high level	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
L	Power supply current	XIII	S	100	344	100	344	100	344	100	344	100	344	mA
M	Power supply transient rise time	XIV	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs
N1-N16	Transition time (t_{TLH}), C_L = 100 pF	XV	S	5	45	5	50	5	50	5	50	5	50	ns
P1-P16	Transition time (t_{THL}), C_L = 100 pF	XVI	S	5	20	5	25	5	25	4	25	4	25	ns
Q1-Q20	Propagation delay time (t_{PHL}) and minimum parallel and serial setup time check; C_L = 100 pF	XVII	S	5	55	5	60	5	60	5	60	5	60	ns
R1-R20	Propagation delay time (t_{PLH}) and minimum parallel and serial setup time check; C_L = 100 pF	XVIII	S	5	45	5	50	5	50	5	50	5	50	ns
S1-S20	Parallel and serial hold time check	XIX	S	5	45	5	50	5	50	5	50	5	50	ns
T1-T4	Maximum shift frequency check and minimum clock pulse width check	XX	S	135	150	135	150	135	150	135	150	135	150	ns
U1-U8	Parallel enable minimum setup and hold time check	XXI	S	5	45	5	50	5	50	5	50	5	50	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V1-V16	Output crosstalk, low level	XXII	Q	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V
V17-V32	Output crosstalk, low level	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W16	Output crosstalk, high level	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XXIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y1-Y4	Input capacitance	XXIV	Q	2	30	2	30	2	30	2	30	2	30	pF
Y5-Y20	Input capacitance	XXIV	Q	2	15	2	15	2	15	2	15	2	15	pF
Y21	Input capacitance	XXIV	Q	2	60	2	60	2	60	2	60	2	60	pF
Y22	Input capacitance	XXIV	Q	2	60	2	60	2	60	2	60	2	60	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, C, or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, G where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/112 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658703. Part number M28787/112-1 supersedes part numbers 2658703 and M28787/112.

6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Clock triggering edge	Positive	Positive	---
Supply voltage	5 ±10%	5 ±10%	V dc
Maximum power dissipation (outputs not loaded)	1.9	1.9	W
Shift frequency	DC to 15	DC to 15	MHz
Clock pulse width	35 min	35 min	ns
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2 min, V_{CC} max	2 min, V_{CC} max	V
Input current (parallel inputs)			
Low level (V_{IL} = 0.0 V)	-0.1 min, -1.73 max	-0.1 min, -1.73 max	mA
(V_{IL} = 0.4 V)	-0.1 min, -1.60 max	-0.1 min, -1.60 max	mA
High level (V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
Input current (serial inputs)			
Low level (V_{IL} = 0.0 V)	-0.10 min, -3.46 max	-0.10 min, -3.46 max	mA
(V_{IL} = 0.4 V)	-0.1 min, -3.2 max	-0.1 min, -3.2 max	mA
High level (V_{IH} = 2.4 V)	0 min, 80 max	0 min, 80 max	µA
(V_{IH} = 5.5 V)	0 min, 2 max	0 min, 2 max	mA
Input current (clock input)			
Low level (V_{IL} = 0.0 V)	-1.0 min, -13.8 max	-1.0 min, -13.8 max	mA
(V_{IL} = 0.4 V)	-1.0 min, -12.8 max	-1.0 min, -12.8 max	mA
High level (V_{IH} = 2.4 V)	0 min, 320 max	0 min, 320 max	µA
(V_{IH} = 5.5 V)	0 min, 4 max	0 min, 4 max	mA
Input current (parallel enable input)			
Low level (V_{IL} = 0.0 V)	-1.0 min, -15.9 max	-1.0 min, -15.9 max	mA
(V_{IL} = 0.4 V)	-1.0 min, -14.8 max	-1.0 min, -14.8 max	mA
High level (V_{IH} = 2.4 V)	0 min, 368 max	0 min, 368 max	µA
(V_{IH} = 5.5 V)	0 min, 4 max	0 min, 4 max	mA
Output voltage			
Low level (I_{OL} = 9.6 mA)	0.0 min, 0.4 max	0.0 min, 0.4 max	V
High level (I_{OH} = 380 µA)	2.4 min, V_{CC} max	2.4 min, V_{CC} max	V
Output drive current			
Low level (V_{OL} = 0.4 V)	9.6 max	9.6 max	mA
High level (V_{OH} = 2.4 V)	-380 min	-380 min	µA
Propagation delay time, clock to output (C_L = 100 pF)			
t_{PLH}	5 min, 50 max	5 min, 50 max	ns
t_{PHL}	5 min, 60 max	5 min, 60 max	ns
Setup time (before positive transition to clock)			
Parallel enable (t_{SH})	45 min	45 min	ns
Serial-parallel data (t_{SL})	45 min	45 min	ns
Hold time (after positive transition of clock), $\overline{PE}$ to CP			
t_{HH}	-10 min	-10 min	ns
t_{HL}	-10 min	-10 min	ns
Hold time (after positive transition of clock), SER, PA, PB, PC, PD to CP			
t_{HH}	0 min	0 min	ns
t_{HL}	0 min	0 min	ns

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* 6.6 Module function. This module contains four 4-bit serial/parallel shift registers. Each 4-bit shift register has a parallel enable input, serial input, four parallel inputs, and four corresponding parallel outputs. All four shift registers are controlled by one master parallel enable and one master clock input. When the parallel enable input is low, the next condition of the shift register synchronous with the clock input is determined by the parallel inputs. When the parallel enable is high, the shift register performs a one bit shift for each clock transition low to high. Data on the serial input is shifted into the QA output and the QA is shifted into the QB, the QB is shifted into the QC, the QC is shifted into the QD, and the QD is dropped. With four low-to-high transitions of the clock, the data on the serial input is completely shifted through the register and dropped.

6.6.1 Input to output relationship. Each 4-bit shift register has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 6/

Parallel enable	Parallel inputs				Serial input	Outputs							
						t_n				$t_n + 1$			
PE	PA	PB	PC	PD	SER	QA	QB	QC	QD	QA	QB	QC	QD
L	PA	PB	PC	PD	X	X	X	X	X	PA	PB	PC	PD
H	X	X	X	X	H	PA	PB	PC	PD	H	PA	PB	PC
H	X	X	X	X	H	H	PA	PB	PC	H	H	PA	PB
H	X	X	X	X	H	H	H	PA	PB	H	H	H	PA
H	X	X	X	X	H	H	H	H	PA	H	H	H	H
H	X	X	X	X	L	PA	PB	PC	PD	L	PA	PB	PC
H	X	X	X	X	L	L	PA	PB	PC	L	L	PA	PB
H	X	X	X	X	L	L	L	PA	PB	L	L	L	PA
H	X	X	X	X	L	L	L	L	PA	L	L	L	L

1/ t_n = Output state at time prior to low-to-high transition of clock.

2/ $t_n + 1$ = Output state at time after low-to-high transition of clock.

3/ X = Don't care term.

4/ PA, PB, PC, PD = Known input (either high level or low level).

5/ L = Low level.

6/ H = High level.

6.6.2 Parallel outputs. Paralleling of outputs is not permissible.

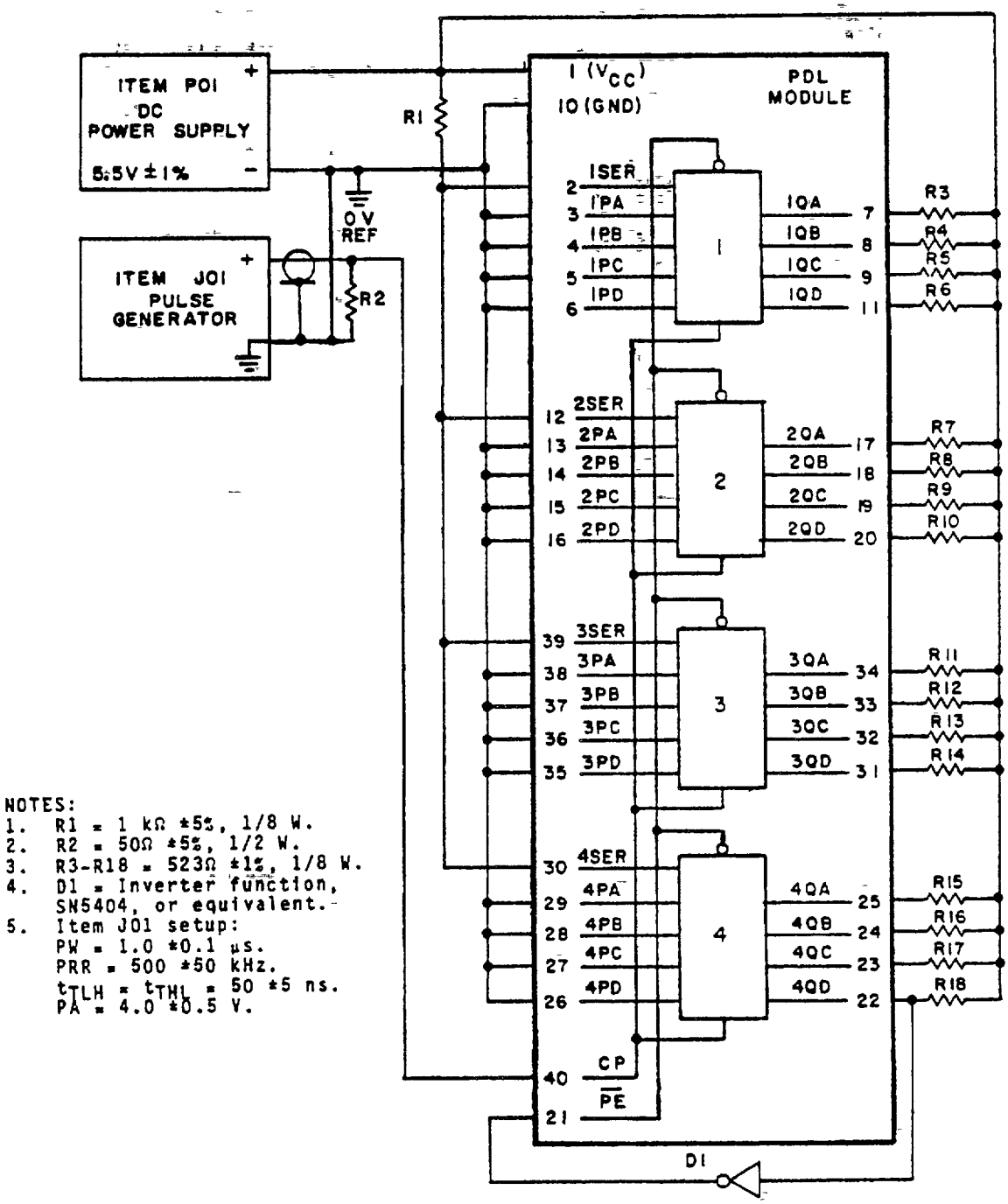
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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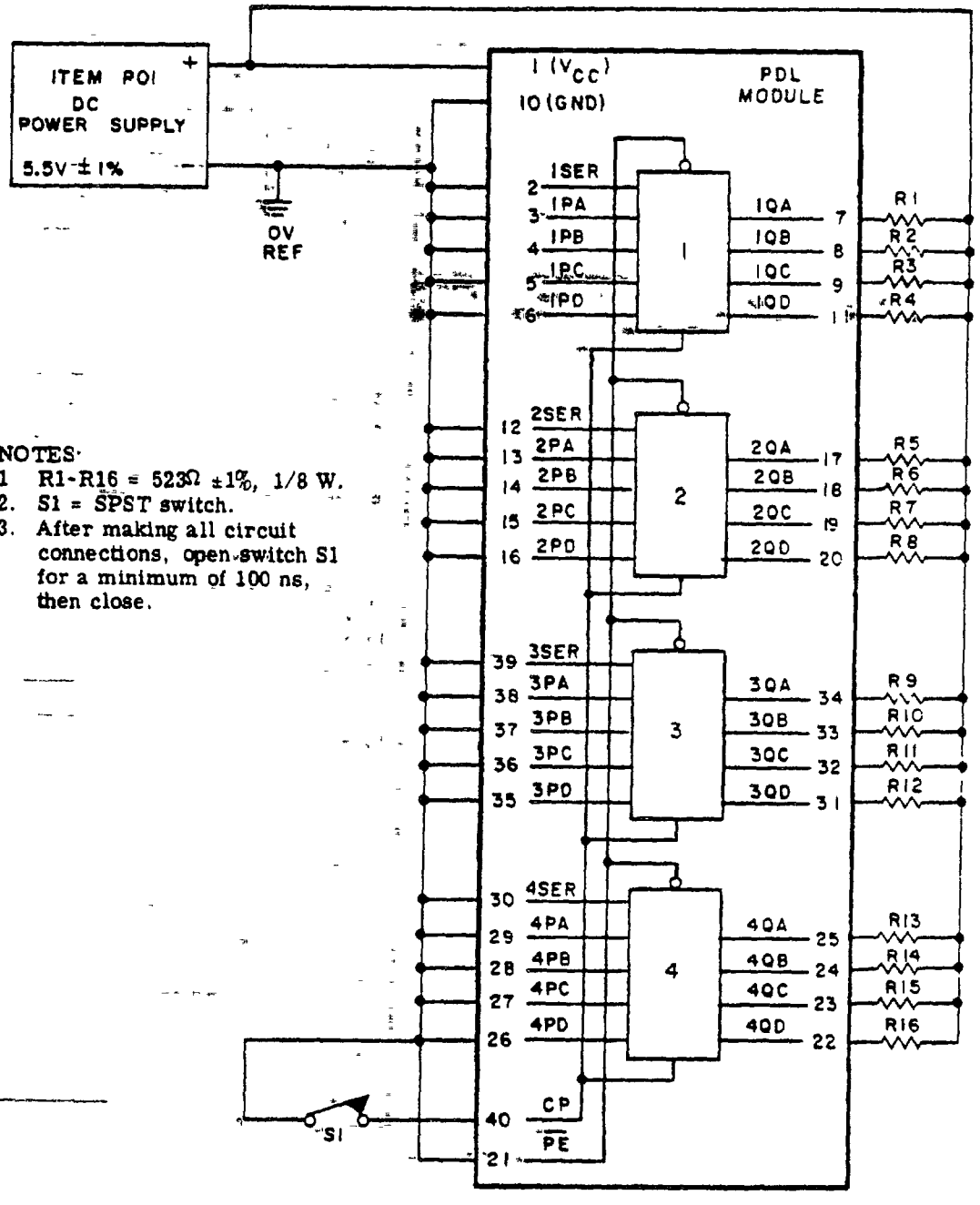
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NOTES:
1. R1 = 1 kΩ ±5%, 1/8 W.
2. R2 = 50Ω ±5%, 1/2 W.
3. R3-R18 = 523Ω ±1%, 1/8 W.
4. D1 = Inverter function, SN5404, or equivalent.
5. Item J01 setup:
   PW = 1.0 ±0.1 µs.
   PRR = 500 ±50 kHz.
   $t_{TLH} = t_{THL}$ = 50 ±5 ns.
   PA = 4.0 ±0.5 V.

* FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R16 = 523Ω ±1%, 1/8 W.
2. S1 = SPST switch.
3. After making all circuit connections, open switch S1 for a minimum of 100 ns, then close.

FIGURE 3. Component temperature test circuit.

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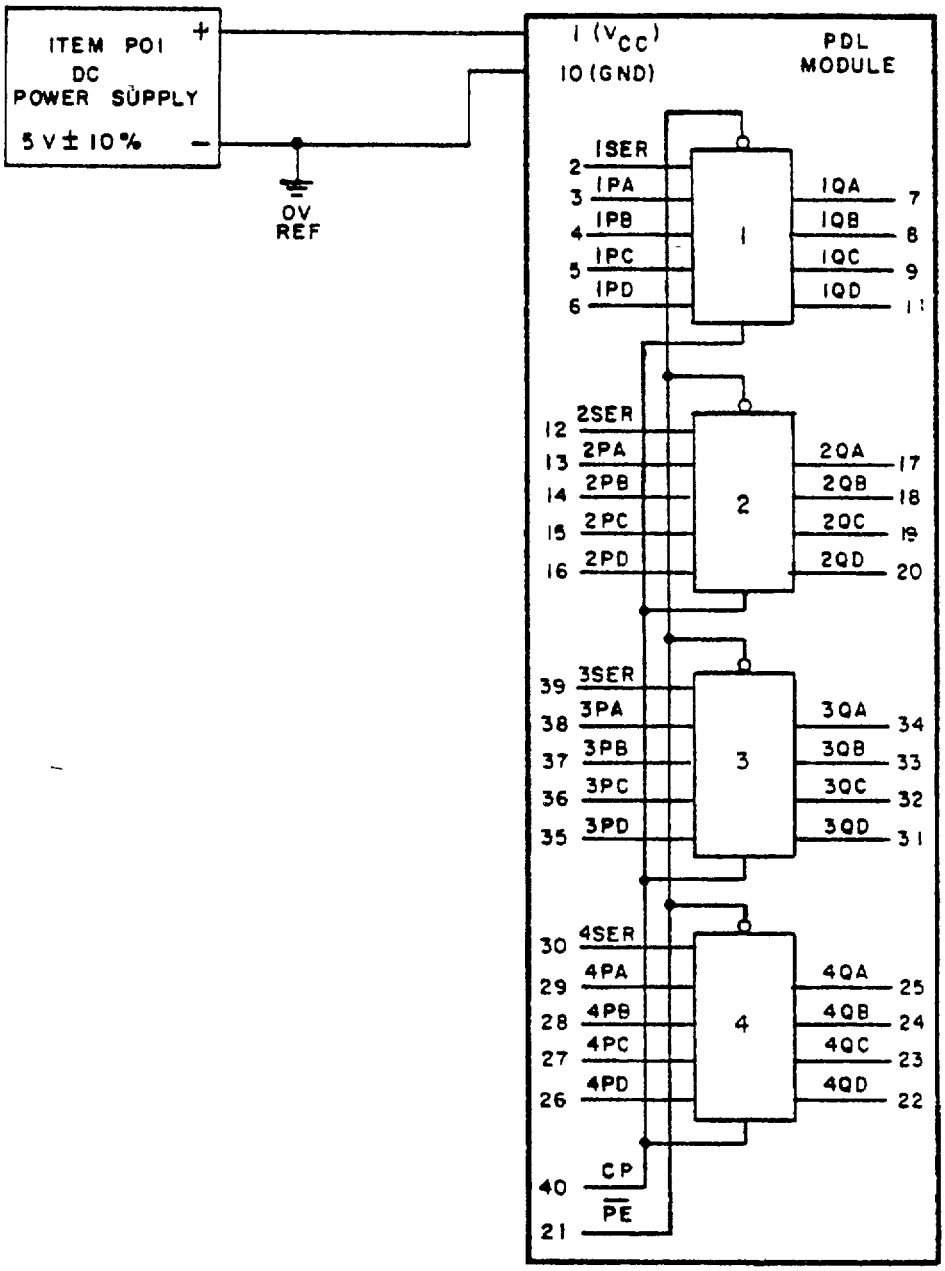


FIGURE 4. Vibration test circuit.

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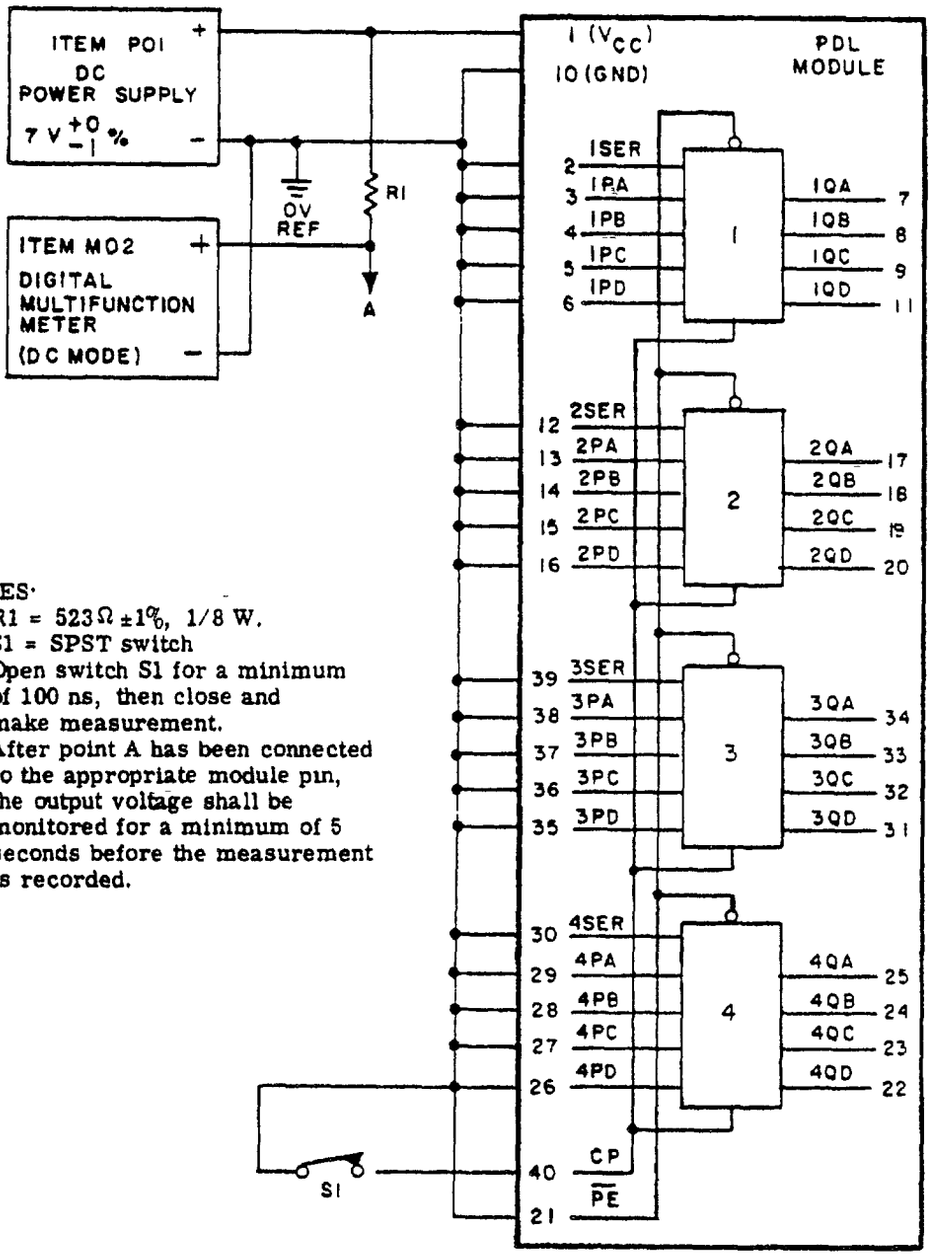
* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1QA V_{OUT} (low loaded)	7	10	800	mV	5
A2	1QB "	8	"	"	"	"
A3	1QC "	9	"	"	"	"
A4	1QD "	11	"	"	"	"
A5	2QA "	17	"	"	"	"
A6	2QB "	18	"	"	"	"
A7	2QC "	19	"	"	"	"
A8	2QD "	20	"	"	"	"
A9	3QA "	34	"	"	"	"
A10	3QB "	33	"	"	"	"
A11	3QC "	32	"	"	"	"
A12	3QD "	31	"	"	"	"
A13	4QA "	25	"	"	"	"
A14	4QB "	24	"	"	"	"
A15	4QC "	23	"	"	"	"
A16	4QD "	22	"	"	"	"

TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Isolation	C	Frame	10	∞	MΩ	6

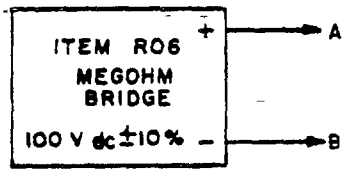
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NOTES:
1. R1 = 523Ω ±1%, 1/8 W.
2. S1 = SPST switch
3. Open switch S1 for a minimum of 100 ns, then close and make measurement.
4. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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WARNING.

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

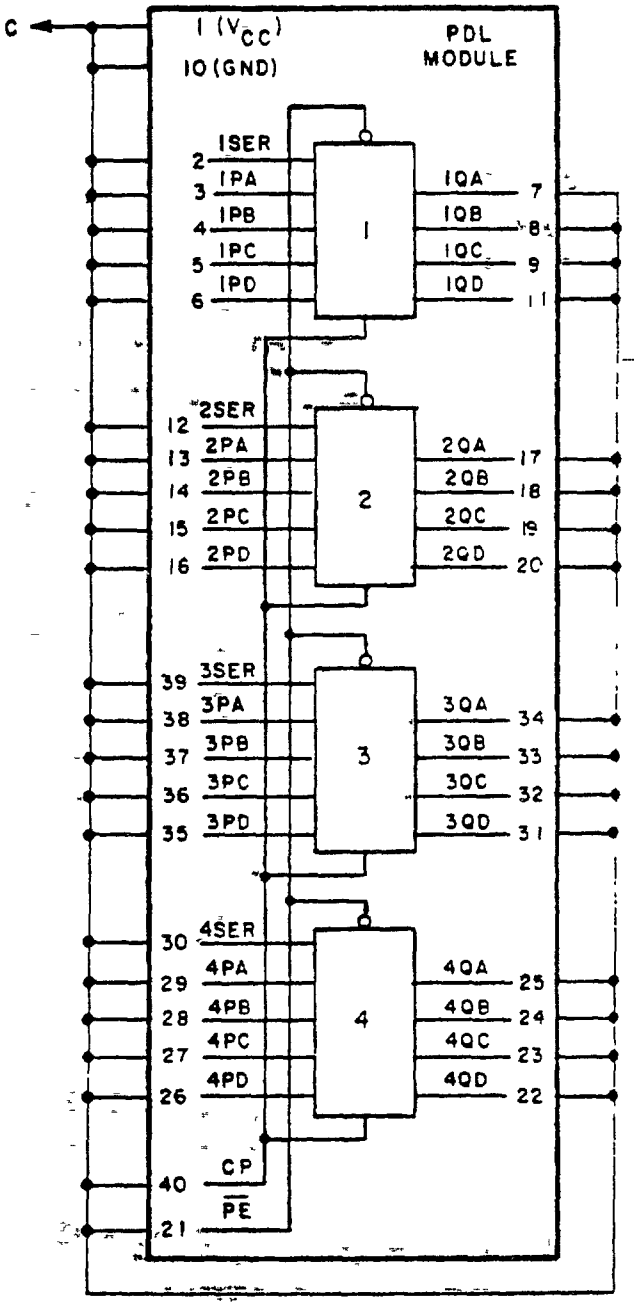


FIGURE 6. Isolation test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1SER	Input breakdown current	3-6,12-16,21,26-30,35-40	2	0	2	mA	7
C2	2SER	"	2-6,13-16,21,26-30,35-40	12	"	"	"	"
C3	3SER	"	2-6,12-16,21,26-30,35-38,40	39	"	"	"	"
C4	4SER	"	2-6,12-16,21,26-29,35-40	30	"	"	"	"
C5	1PA	"	2,4-6,12-16,26-30,35-40	3	"	1	"	"
C6	1PB	"	2,3,5,6,12-16,26-30,35-40	4	"	"	"	"
C7	1PC	"	2-4,6,12-16,26-30,35-40	5	"	"	"	"
C8	1PD	"	2-5,12-16,26-30,35-40	6	"	"	"	"
C9	2PA	"	2-6,12,14-16,26-30,35-40	13	"	"	"	"
C10	2PB	"	2-6,12,13,15,16,26-30,35-40	14	"	"	"	"
C11	2PC	"	2-6,12-14,16,26-30,35-40	15	"	"	"	"
C12	2PD	"	2-6,12-15,26-30,35-40	16	"	"	"	"
C13	3PA	"	2-6,12-16,26-30,35-37,39,40	38	"	"	"	"
C14	3PB	"	2-6,12-16,26-30,35,36,38-40	37	"	"	"	"
C15	3PC	"	2-6,12-16,26-30,35,37-40	36	"	"	"	"
C16	3PD	"	2-6,12-16,26-30,36-40	35	"	"	"	"
C17	4PA	"	2-6,12-16,26-28,30,35-40	29	"	"	"	"
C18	4PB	"	2-6,12-16,26,27,29,30,35-40	28	"	"	"	"
C19	4PC	"	2-6,12-16,26,28-30,35-40	27	"	"	"	"
C20	4PD	"	2-6,12-16,27-30,35-40	26	"	"	"	"
C21	CP	"	2-6,12-16,26-30,35-39,21	40	"	4	"	"
C22	$\overline{PE}$	"	2-6,12-16,26-30,35-40	21	"	4	"	"

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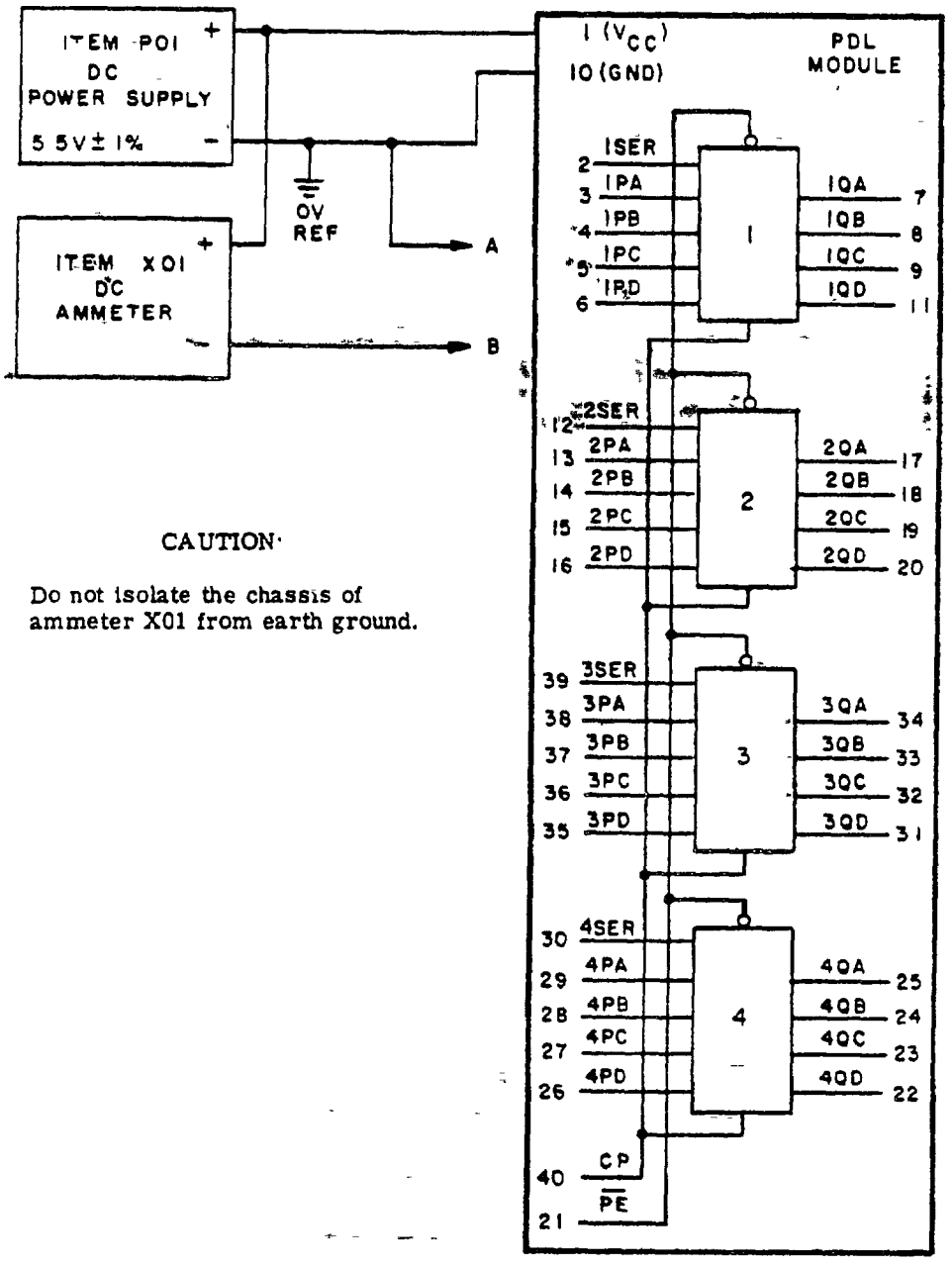


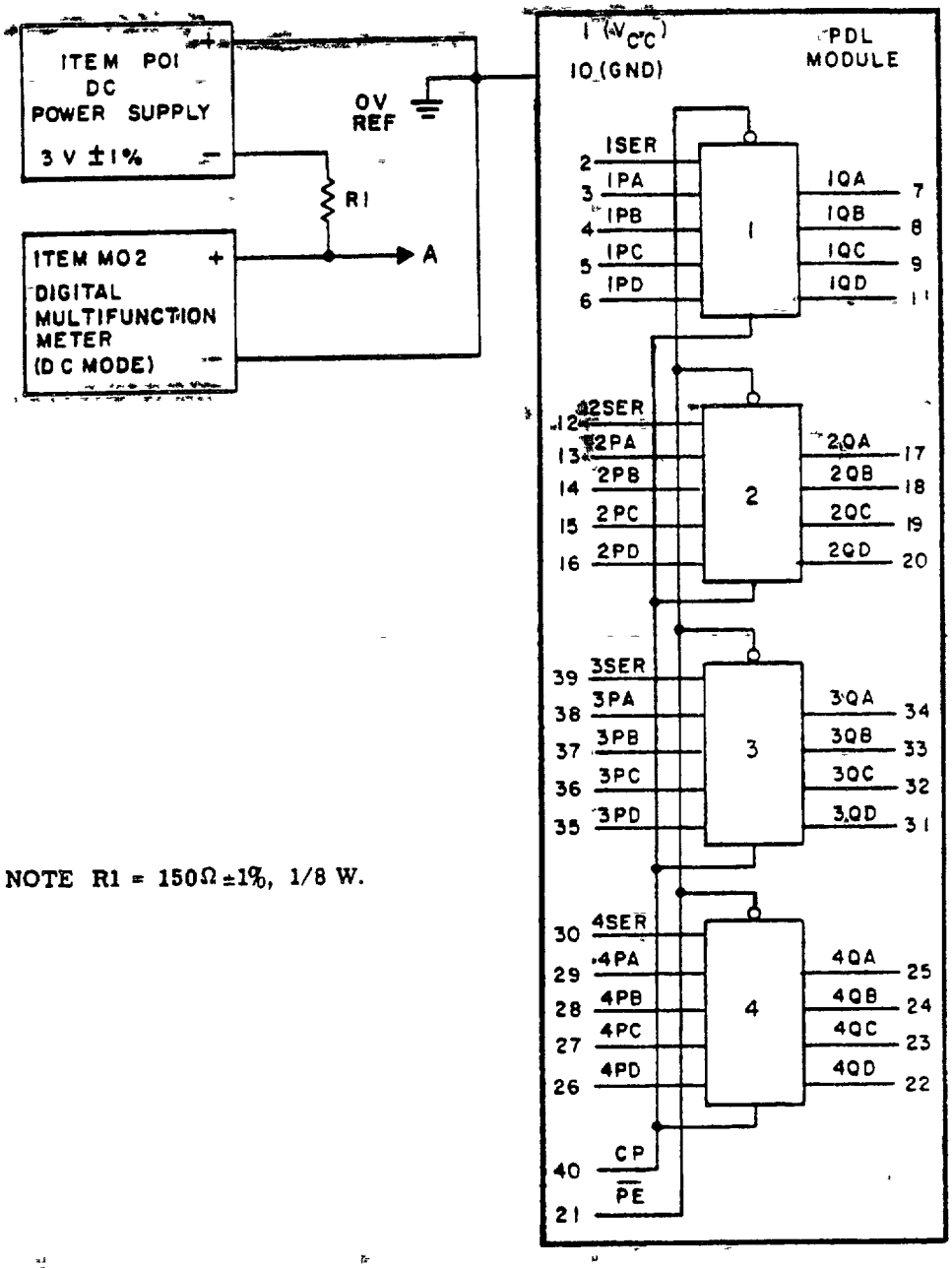
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1SER	Input clamp diode voltage	2	-0.4	-1.5	V	8
D2	2SER	"	12	"	"	"	"
D3	3SER	"	39	"	"	"	"
D4	4SER	"	30	"	"	"	"
D5	1PA	"	3	"	"	"	"
D6	1PB	"	4	"	"	"	"
D7	1PC	"	5	"	"	"	"
D8	1PD	"	6	"	"	"	"
D9	2PA	"	13	"	"	"	"
D10	2PB	"	14	"	"	"	"
D11	2PC	"	15	"	"	"	"
D12	2PD	"	16	"	"	"	"
D13	3PA	"	38	"	"	"	"
D14	3PB	"	37	"	"	"	"
D15	3PC	"	36	"	"	"	"
D16	3PD	"	35	"	"	"	"
D17	4PA	"	29	"	"	"	"
D18	4PB	"	28	"	"	"	"
D19	4PC	"	27	"	"	"	"
D20	4PD	"	26	"	"	"	"
D21	CP	"	40	"	"	"	"
D22	$\overline{PE}$	"	21	"	"	"	"

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NOTE R1 = 150Ω ±1%, 1/8 W.

* FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1SER Input current, high level	2	Open	Open	21	0	80	µA	9
E2	2SER "	12	"	"	"	"	"	"	"
E3	3SER "	39	"	"	"	"	"	"	"
E4	4SER "	30	"	"	"	"	"	"	"
E5	1PA "	3	"	"	Open	"	40	"	"
E6	1PB "	4	"	"	"	"	"	"	"
E7	1PC "	5	"	"	"	"	"	"	"
E8	1PD "	6	"	"	"	"	"	"	"
E9	2PA "	13	"	"	"	"	"	"	"
E10	2PB "	14	"	"	"	"	"	"	"
E11	2PC "	15	"	"	"	"	"	"	"
E12	2PD "	16	"	"	"	"	"	"	"
E13	3PA "	38	"	"	"	"	"	"	"
E14	3PB "	37	"	"	"	"	"	"	"
E15	3PC "	36	"	"	"	"	"	"	"
E16	3PD "	35	"	"	"	"	"	"	"
E17	4PA "	29	"	"	"	"	"	"	"
E18	4PB "	28	"	"	"	"	"	"	"
E19	4PC "	27	"	"	"	"	"	"	"
E20	4PD "	26	"	"	"	"	"	"	"
E21	CP "	40	"	"	"	"	320	"	"
E22	$\overline{PE}$ "	21	"	"	"	"	368	"	"

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TABLE VIII. Input current (low-level) test.

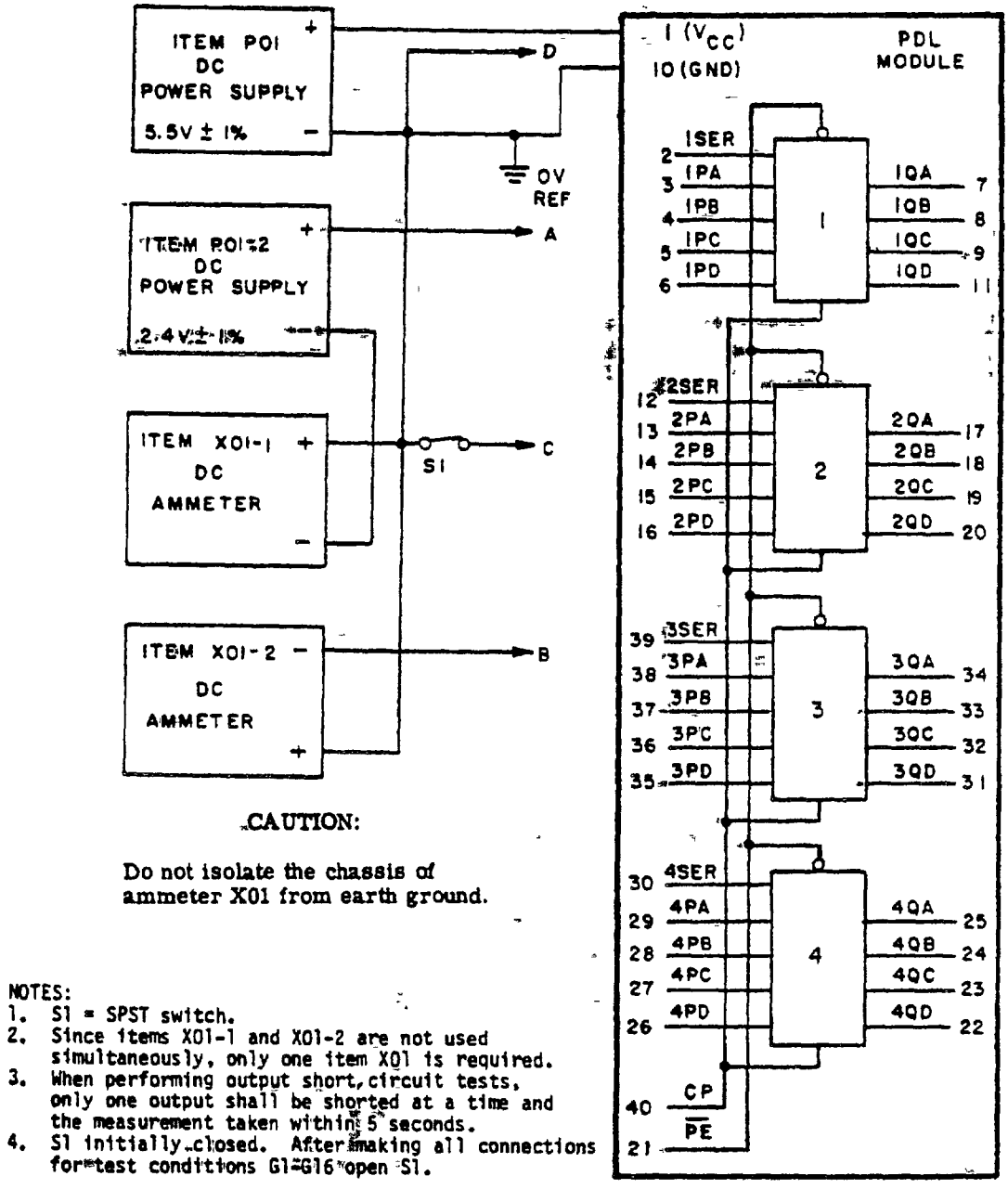
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1SER Input current, low level	21	2	Open	Open	-0.10	-3.46	mA	9
F2	2SER "	"	12	"	"	"	"	"	"
F3	3SER "	"	39	"	"	"	"	"	"
F4	4SER "	"	30	"	"	"	"	"	"
F5	1PA "	Open	3	"	21	"	-1.73	"	"
F6	1PB "	"	4	"	"	"	"	"	"
F7	1PC "	"	5	"	"	"	"	"	"
F8	1PD "	"	6	"	"	"	"	"	"
F9	2PA "	"	13	"	"	"	"	"	"
F10	2PB "	"	14	"	"	"	"	"	"
F11	2PC "	"	15	"	"	"	"	"	"
F12	2PD "	"	16	"	"	"	"	"	"
F13	3PA "	"	38	"	"	"	"	"	"
F14	3PB "	"	37	"	"	"	"	"	"
F15	3PC "	"	36	"	"	"	"	"	"
F16	3PD "	"	35	"	"	"	"	"	"
F17	4PA "	"	29	"	"	"	"	"	"
F18	4PB "	"	28	"	"	"	"	"	"
F19	4PC "	"	27	"	"	"	"	"	"
F20	4PD "	"	26	"	"	"	"	"	"
F21	CP "	"	40	"	Open	-1.0	-13.8	"	"
F22	$\overline{PE}$ "	"	21	"	Open	-1.0	-15.9	"	"

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TABLE IX. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1QA Output short circuit current	Open	7	40	21	-10	-80	mA	9
G2	1QB "	"	8	"	"	"	"	"	"
G3	1QC "	"	9	"	"	"	"	"	"
G4	1QD "	"	11	"	"	"	"	"	"
G5	2QA "	"	17	"	"	"	"	"	"
G6	2QB "	"	18	"	"	"	"	"	"
G7	2QC "	"	19	"	"	"	"	"	"
G8	2QD "	"	20	"	"	"	"	"	"
G9	3QA "	"	34	"	"	"	"	"	"
G10	3QB "	"	33	"	"	"	"	"	"
G11	3QC "	"	32	"	"	"	"	"	"
G12	3QD "	"	31	"	"	"	"	"	"
G13	4QA "	"	25	"	"	"	"	"	"
G14	4QB "	"	24	"	"	"	"	"	"
G15	4QC "	"	23	"	"	"	"	"	"
G16	4QD "	"	22	"	"	"	"	"	"

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NOTES:
1. S1 = SPST switch.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
3. When performing output short-circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.
4. S1 initially closed. After making all connections for test conditions G1-G16 open S1.

FIGURE 9. Input current (high-level and low-level) and output short circuit current test circuit for tables VII, VIII, and IX.

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* TABLE X. Functional shift right (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Functional shift right (input threshold clock), low level	3-6	21	2	Open	See note 4, figure 10		- -	10
H2	"	13-16	"	12	"	"		"	"
H3	"	35-38	"	39	"	"		"	"
H4	"	26-29	"	30	"	"		"	"
H5	Functional shift right (input threshold clock), high level	2	"	3-6	"	See note 5, figure 10		"	"
H6	"	12	"	13-16	"	"		"	"
H7	"	39	"	35-38	"	"		"	"
H8	"	30	"	26-29	"	"		"	"

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* TABLE XI. Output voltage (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1QA Output voltage, (input threshold conditions), parallel, low-level	2,4-6	Open	3,21	7	0	400	mV	10
J2	1QB "	2,3,5,6	"	4,21	8	"	"	"	"
J3	1QC "	2-4,6	"	5,21	9	"	"	"	"
J4	1QD "	2-5	"	6,21	11	"	"	"	"
J5	2QA "	12,14-16	"	13,21	17	"	"	"	"
J6	2QB "	12,13, 15,16	"	14,21	18	"	"	"	"
J7	2QC "	12-14,16	"	15,21	19	"	"	"	"
J8	2QD "	12-15	"	16,21	20	"	"	"	"
J9	3QA "	35-37,39	"	21,38	34	"	"	"	"
J10	3QB "	35,36, 38,39	"	21,37	33	"	"	"	"
J11	3QC "	35,37-39	"	21,36	32	"	"	"	"
J12	3QD "	36-39	"	21,35	31	"	"	"	"
J13	4QA "	26-28,30	"	21,29	25	"	"	"	"
J14	4QB "	26,27, 29,30	"	21,28	24	"	"	"	"
J15	4QC "	26,28-30	"	21,27	23	"	"	"	"
J16	4QD "	27-30	"	21,26	22	"	"	"	"

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* TABLE XI. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J17	1QA Output voltage, (input threshold conditions), serial, low level	3-6	21	2	7	0	400	mV	10
J18	2QA "	13-16	"	12	17	"	"	"	"
J19	3QA "	35-38	"	39	34	"	"	"	"
J20	4QA "	26-29	"	30	25	"	"	"	"

* TABLE XII. Output voltage (high-level) test.

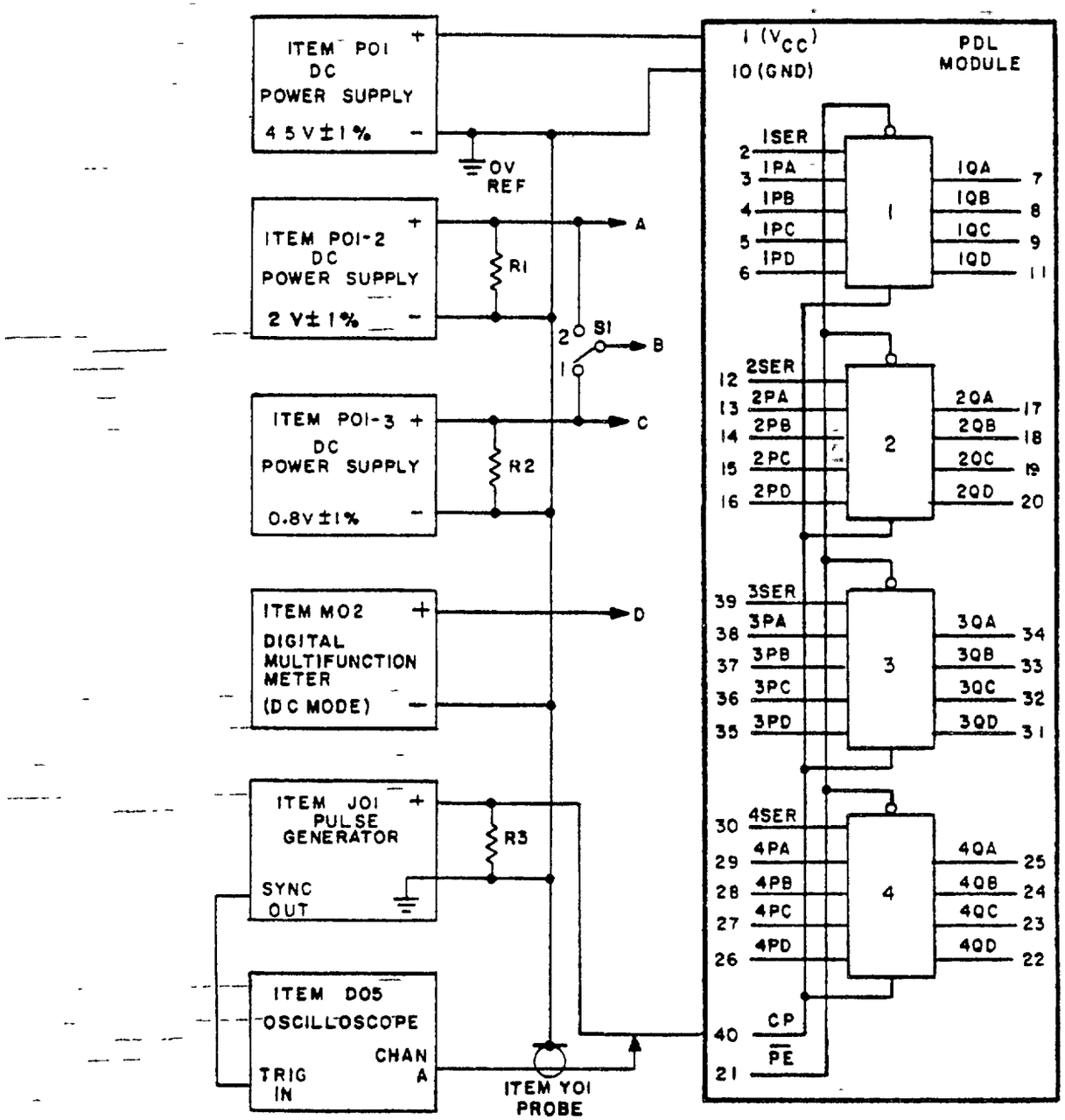
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	1QA Output voltage, (input threshold conditions), parallel, high level	3	Open	2,4-6,21	7	2.4	4.5	V	10
K2	1QB "	4	"	2,3,5,6,21	8	"	"	"	"
K3	1QC "	5	"	2-4,6,21	9	"	"	"	"
K4	1QD "	6	"	2-5,21	11	"	"	"	"
K5	2QA "	13	"	12,14-16,21	17	"	"	"	"
K6	2QB "	14	"	12,13,15,16,21	18	"	"	"	"

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* TABLE XII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K7	2QC Output voltage, (input threshold conditions), parallel, high level	15	Open	12-14, 16,21	19	2.4	4.5	V	10
K8	2QD "	16	"	12-15,21	20	"	"	"	"
K9	3QA "	38	"	21,35-37,39	34	"	"	"	"
K10	3QB "	37	"	21,35, 36,38, 39	33	"	"	"	"
K11	3QC "	36	"	21,35, 37-39	32	"	"	"	"
K12	3QD "	35	"	21,36-39	31	"	"	"	"
K13	4QA "	29	"	21,26-28,30	25	"	"	"	"
K14	4QB "	28	"	21,26, 27,29, 30	24	"	"	"	"
K15	4QC "	27	"	21,26, 28-30	23	"	"	"	"
K16	4QD "	26	"	21,27-30	22	"	"	"	"
K17	1QA Output voltage, (input threshold conditions), serial, high level	2	21	3-6	7	"	"	"	"
K18	2QA "	12	"	13-16	17	"	"	"	"
K19	3QA "	39	"	35-38	34	"	"	"	"
K20	4QA "	30	"	26-29	25	"	"	"	"

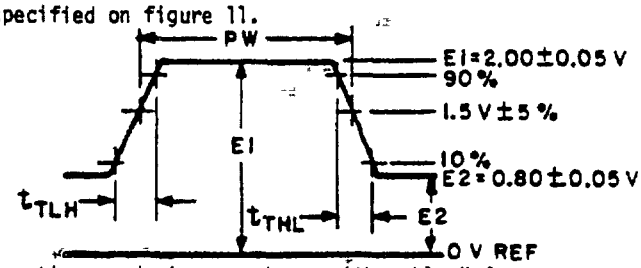
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* FIGURE 10. Functional shift right (low-level and high-level) and output voltage (low-level and high-level) test circuit for tables X, XI, and XII.

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NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1/2 W.
2. S1 = SPDT switch.
3. Load all the outputs with a load as specified on figure 11.
4. Test condition H1-H4:
   a. Clock pulse:
      PW = 500 ±50 ns.
      $t_{TLH} = t_{THL}$ = 100 ±10 ns.



   b. Set up the pulse using a repetition rate of 1 MHz ±10% then place item J01 in the single pulse mode.
   c. With S1 in position 1 and all connections made in accordance with table X for a test condition, apply one pulse to the circuit under test. Verify the output states to be:

QA	QB	QC	QD
H	H	H	H

   d. Place S1 in position 2 and apply four pulses to the circuit under test. After each pulse is applied, verify that the output states are as follows:

Number of pulses	QA	QB	QC	QD
1	L	H	H	H
2	L	L	H	H
3	L	L	L	H
4	L	L	L	L

5. Test conditions H5-H8.
   a. The clock pulse shall be as specified in note 4(a) above.
   b. With S1 in position 1 and all connections made in accordance with table X for a test condition, apply one pulse to the circuit under test. Verify the output states to be:

QA	QB	QC	QD
L	L	L	L

* FIGURE 10. Functional shift right (low-level and high-level) and output voltage (low-level and high-level) test circuit for tables X, XI, and XII - Continued.

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NOTES: - Continued.

c. Place S1 in position 2 and apply four pulses to the circuit under test. After each pulse is applied, verify that the output states are as follows:

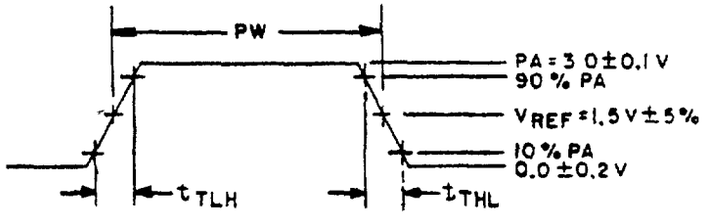
Number of pulses	QA	QB	QC	QD
1	H	L	L	L
2	H	H	L	L
3	H	H	H	L
4	H	H	H	H

6. For test conditions H1-H8, the output logic states on the circuit under test must be detected simultaneously. The total load applied to each output under test (including the load circuit and logic state detecting device) must be in accordance with figure 11.

7. Test conditions J1-J16 and K1-K16:

a. Clock pulse:
PW = 500 ±50 ns.
$t_{TLH} = t_{THL} = 100 \pm 10$ ns.

6. Set up the pulse using a repetition rate of 1 MHz ±10% then place item J01 in the single pulse mode.



c. With all connections made in accordance with tables XI and XII for a test condition, apply one pulse to the circuit under test. Make measurement of QA, QB, QC, QD.

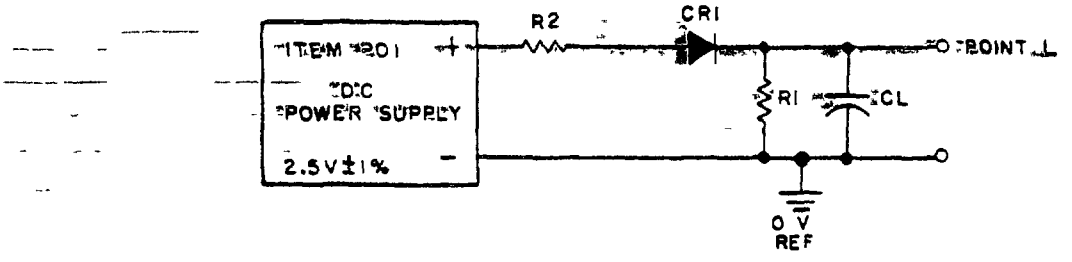
8. Test conditions J17-J20 and K17-K20:

a. The clock pulse shall be as specified in note 7(a) above.

b. With S1 in position 1 and all connections made in accordance with tables XI and XII for a test condition, apply one pulse to the circuit under test. Place S1 in position 2 and apply one pulse to the circuit under test. Make measurement QA.

* FIGURE 10. Functional shift right (low-level and high-level) and output voltage (low-level and high-level) test circuit for tables X, XI, and XII - Continued.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = 6.25 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 9.6 mA ±5% out of point L with point L held at 0.4 V.
4. For test conditions V and W, C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. For test conditions N, P, Q, R, S, T and U, C_L = 100 pF ±5 pF (including probe and fixture capacitance).
6. Capacitor C_L is not required for dc electrical test conditions H1-H8, J1-J20, K1-K20.

FIGURE 11. PDL logic load circuit.

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TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	100	344	mA	12

TABLE XIV. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply transient rise time	2.5	7.5	µs	13,14

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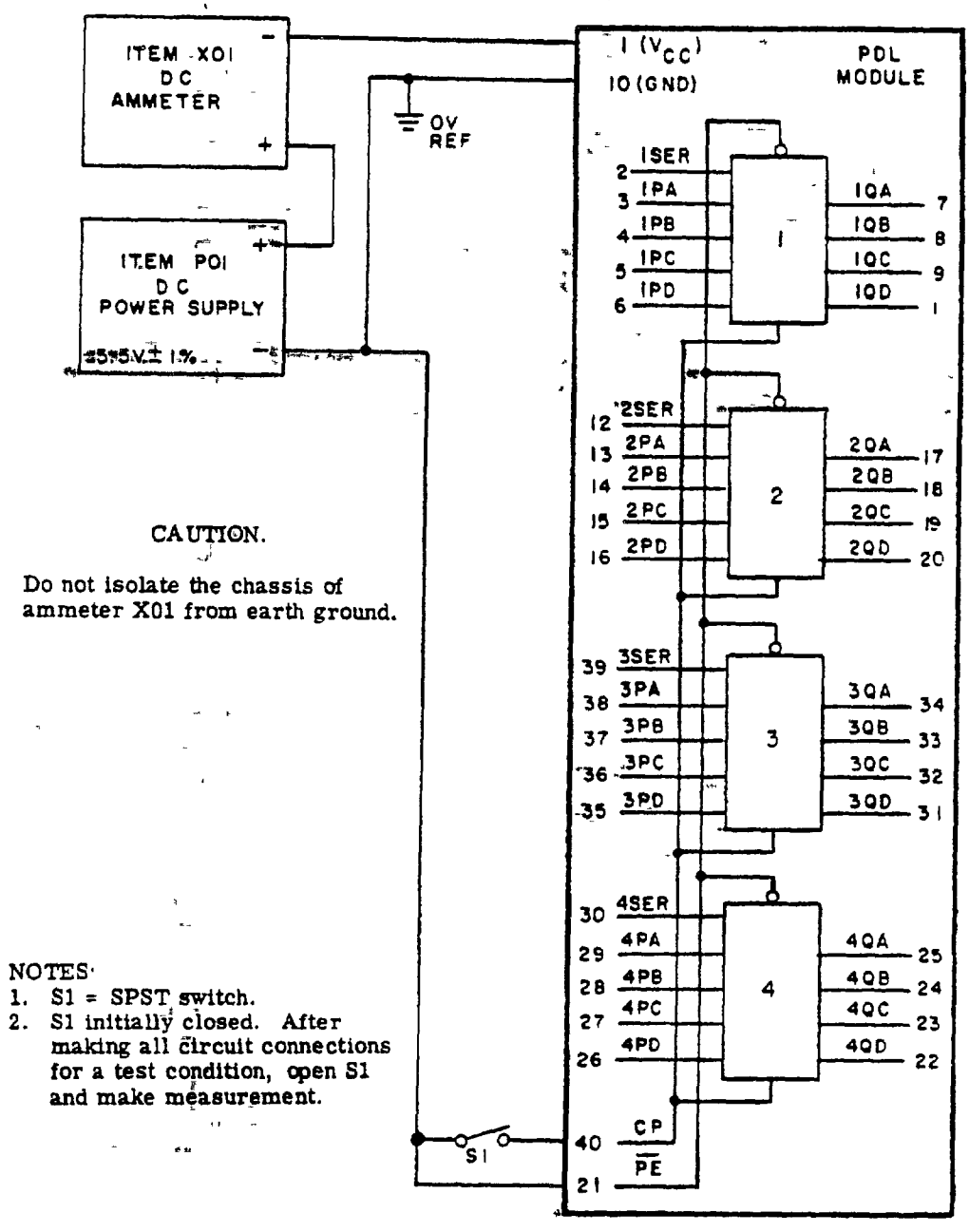
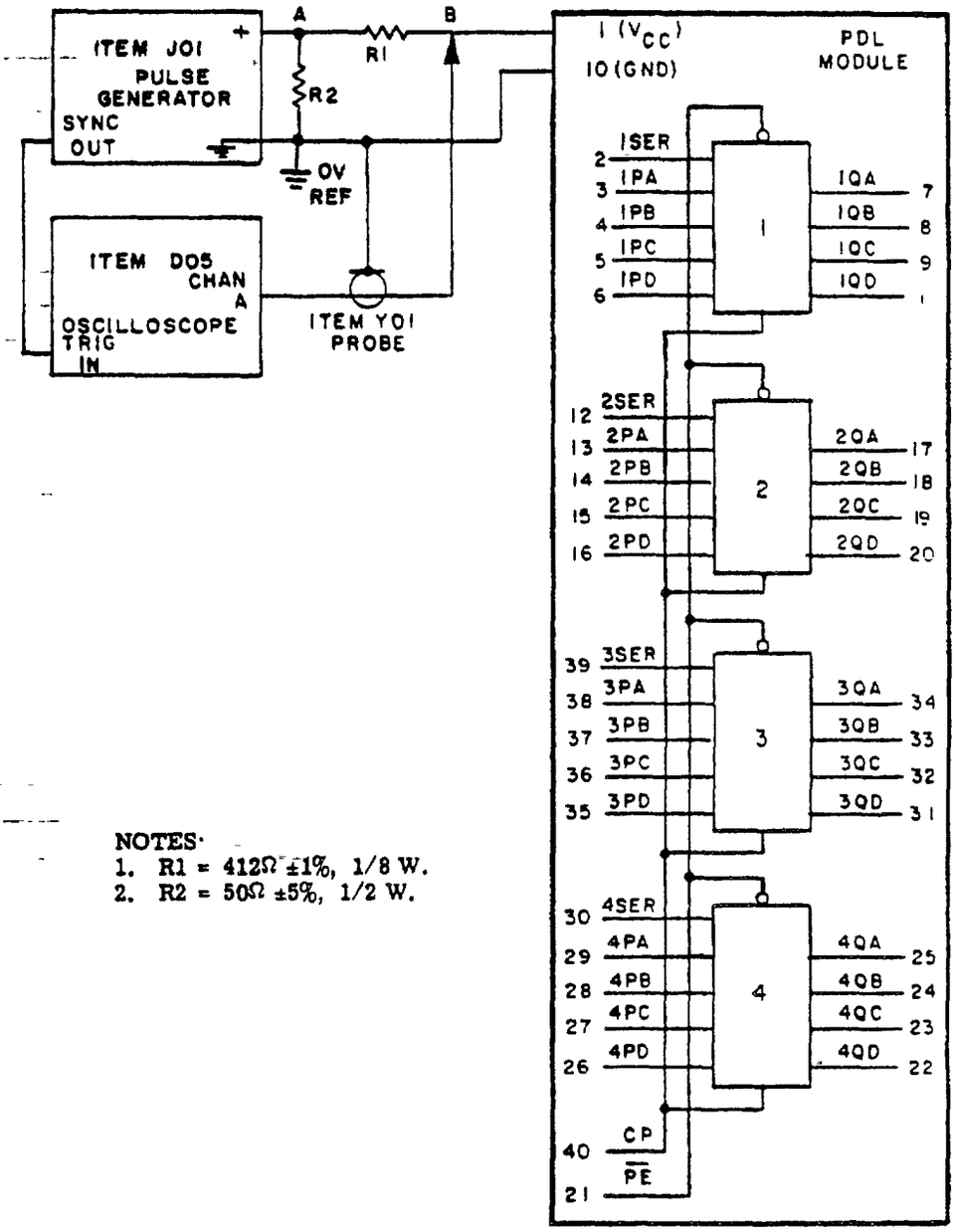


FIGURE 12. Power supply current test circuit for table XIII.

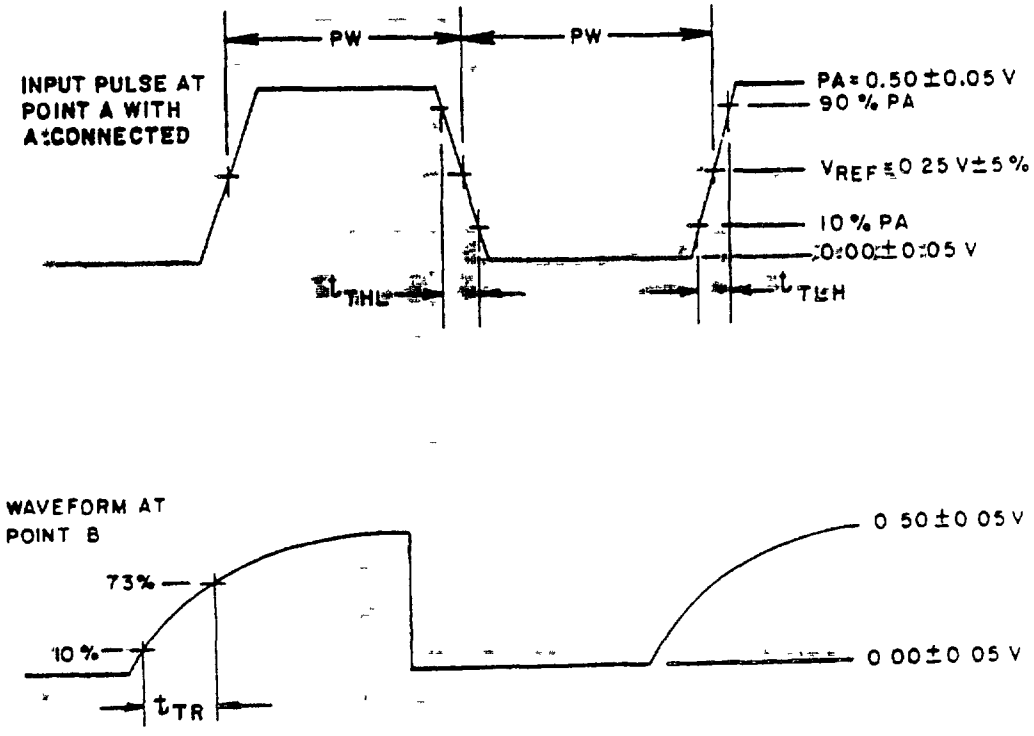
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1/2 W.

FIGURE 13. Power supply transient rise time test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   PW = 30 ±3 µs.
   $t_{TLH} = t_{THL} = 50 \pm 5$ ns.
2. Measure t_{TR} at point B.

★ FIGURE 14. Power supply transient rise time test circuit-waveforms for table XIV.

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* TABLE XV. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	1QA Transition time (t_{TLH}), C_L = 100 pF	2	3-6	40	21	7	5	45	ns	15,16
N2	1QB "	"	"	"	"	8	"	"	"	"
N3	1QC "	"	"	"	"	9	"	"	"	"
N4	1QD "	"	"	"	"	11	"	"	"	"
N5	2QA "	12	13-16	"	"	17	"	"	"	"
N6	2QB "	"	"	"	"	18	"	"	"	"
N7	2QC "	"	"	"	"	19	"	"	"	"
N8	2QD "	"	"	"	"	20	"	"	"	"
N9	3QA "	39	35-38	"	"	34	"	"	"	"
N10	3QB "	"	"	"	"	33	"	"	"	"
N11	3QC "	"	"	"	"	32	"	"	"	"
N12	3QD "	"	"	"	"	31	"	"	"	"
N13	4QA "	30	26-29	"	"	25	"	"	"	"
N14	4QB "	"	"	"	"	24	"	"	"	"
N15	4QC "	"	"	"	"	23	"	"	"	"
N16	4QD "	"	"	"	"	22	"	"	"	"

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* TABLE XVI. Transition time (t_{THL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	1QA Transition time (t_{THL}), C_L = 100 pF	2	3-6	21	40	7	5	20	ns	15,16
P2	1QB "	"	"	"	"	8	"	"	"	"
P3	1QC "	"	"	"	"	9	"	"	"	"
P4	1QD "	"	"	"	"	11	"	"	"	"
P5	2QA "	12	13-16	"	"	17	"	"	"	"
P6	2QB "	"	"	"	"	18	"	"	"	"
P7	2QC "	"	"	"	"	19	"	"	"	"
P8	2QD "	"	"	"	"	20	"	"	"	"
P9	3QA "	39	35-38	"	"	34	"	"	"	"
P10	3QB "	"	"	"	"	33	"	"	"	"
P11	3QC "	"	"	"	"	32	"	"	"	"
P12	3QD "	"	"	"	"	31	"	"	"	"
P13	4QA "	30	26-29	"	"	25	"	"	"	"
P14	4QB "	"	"	"	"	24	"	"	"	"
P15	4QC "	"	"	"	"	23	"	"	"	"
P16	4QD "	"	"	"	"	22	"	"	"	"

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* TABLE XVII. Propagation delay time (t_{PHL}) and minimum parallel and serial setup time check test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	CP to 1QA Propagation delay time (t_{PHL}), C_L = 100 pF (minimum parallel setup time check)	2	3-6	21	40	7	5	55	ns	15,16
Q2	CP to 1QB "	"	"	"	"	8	"	"	"	"
Q3	CP to 1QC "	"	"	"	"	9	"	"	"	"
Q4	CP to 1QD "	"	"	"	"	11	"	"	"	"
Q5	CP to 2QA "	12	13-16	"	"	17	"	"	"	"
Q6	CP to 2QB "	"	"	"	"	18	"	"	"	"
Q7	CP to 2QC "	"	"	"	"	19	"	"	"	"
Q8	CP to 2QD "	"	"	"	"	20	"	"	"	"
Q9	CP to 3QA "	39	35-38	"	"	34	"	"	"	"
Q10	CP to 3QB "	"	"	"	"	33	"	"	"	"
Q11	CP to 3QC "	"	"	"	"	32	"	"	"	"
Q12	CP to 3QD "	"	"	"	"	31	"	"	"	"
Q13	CP to 4QA "	30	26-29	"	"	25	"	"	"	"
Q14	CP to 4QB "	"	"	"	"	24	"	"	"	"
Q15	CP to 4QC "	"	"	"	"	23	"	"	"	"
Q16	CP to 4QD "	"	"	"	"	22	"	"	"	"

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* TABLE XVII. Propagation delay time (t_{PHL}) and minimum parallel and serial setup time check test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q17	CP to 1QA	Propagation delay time (t_{PHL}), C_L = 100 pF (minimum serial setup time check)	3-6,21	2	Open	40	7	5	55	ns	15,16
Q18	CP to 2QA	"	13-16, 21	12	"	"	17	"	"	"	"
Q19	CP to 3QA	"	21,35-38	39	"	"	34	"	"	"	"
Q20	CP to 4QA	"	21,26-29	30	"	"	25	"	"	"	"

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* TABLE XVIII. Propagation delay time (t_{PLH}) and minimum parallel and serial setup time check test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R1	CP to 1QA	Propagation delay time (t_{PLH}), C_L = 100 pF (minimum parallel setup time check)	2	3-6	21	40	7	5	45	ns	15,16
R2	CP to 1QB	"	"	"	"	"	8	"	"	"	"
R3	CP to 1QC	"	"	"	"	"	9	"	"	"	"
R4	CP to 1QD	"	"	"	"	"	11	"	"	"	"
R5	CP to 2QA	"	12	13-16	"	"	17	"	"	"	"
R6	CP to 2QB	"	"	"	"	"	18	"	"	"	"
R7	CP to 2QC	"	"	"	"	"	19	"	"	"	"
R8	CP to 2QD	"	"	"	"	"	20	"	"	"	"
R9	CP to 3QA	"	39	35-38	"	"	34	"	"	"	"
R10	CP to 3QB	"	"	"	"	"	33	"	"	"	"
R11	CP to 3QC	"	"	"	"	"	32	"	"	"	"
R12	CP to 3QD	"	"	"	"	"	31	"	"	"	"
R13	CP to 4QA	"	30	26-29	"	"	25	"	"	"	"
R14	CP to 4QB	"	"	"	"	"	24	"	"	"	"
R15	CP to 4QC	"	"	"	"	"	23	"	"	"	"
R16	CP to 4QD	"	"	"	"	"	22	"	"	"	"

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* TABLE XVIII. Propagation delay time (t_{PHL}) and minimum parallel and serial setup time check test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R17	CP to 1QA Propagation delay time (t_{PLH}). C_L = 100 pF (minimum serial setup time check)	3-6,21	2	Open	40	7	5	45	ns	15,16
R18	CP to 2QA "	13,16-21	12	"	"	17	"	"	"	"
R19	CP to 3QA "	21,35-38	39	"	"	34	"	"	"	"
R20	CP to 4QA "	21,26-29	30	"	"	25	"	"	"	"

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* TABLE XIX. Parallel and serial hold time check test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S1	1QA	Parallel hold time check	2	3-6	21	40	7	5	45	ns	15,17
S2	1QB	"	"	"	"	"	8	"	"	"	"
S3	1QC	"	"	"	"	"	9	"	"	"	"
S4	1QD	"	"	"	"	"	11	"	"	"	"
S5	2QA	"	12	13-16	"	"	17	"	"	"	"
S6	2QB	"	"	"	"	"	18	"	"	"	"
S7	2QC	"	"	"	"	"	19	"	"	"	"
S8	2QD	"	"	"	"	"	20	"	"	"	"
S9	3QA	"	39	35-38	"	"	34	"	"	"	"
S10	3QB	"	"	"	"	"	33	"	"	"	"
S11	3QC	"	"	"	"	"	32	"	"	"	"
S12	3QD	"	"	"	"	"	31	"	"	"	"
S13	4QA	"	30	26-29	"	"	25	"	"	"	"
S14	4QB	"	"	"	"	"	24	"	"	"	"
S15	4QC	"	"	"	"	"	23	"	"	"	"
S16	4QD	"	"	"	"	"	22	"	"	"	"
S17	1QA	Serial hold time check	3-6,21	2	Open	"	7	"	"	"	"
S18	2QA	"	13-16, 21	12	"	"	17	"	"	"	"
S19	3QA	"	21,35-38	39	"	"	34	"	"	"	"
S20	4QA	"	21,26-29	30	"	"	25	"	"	"	"

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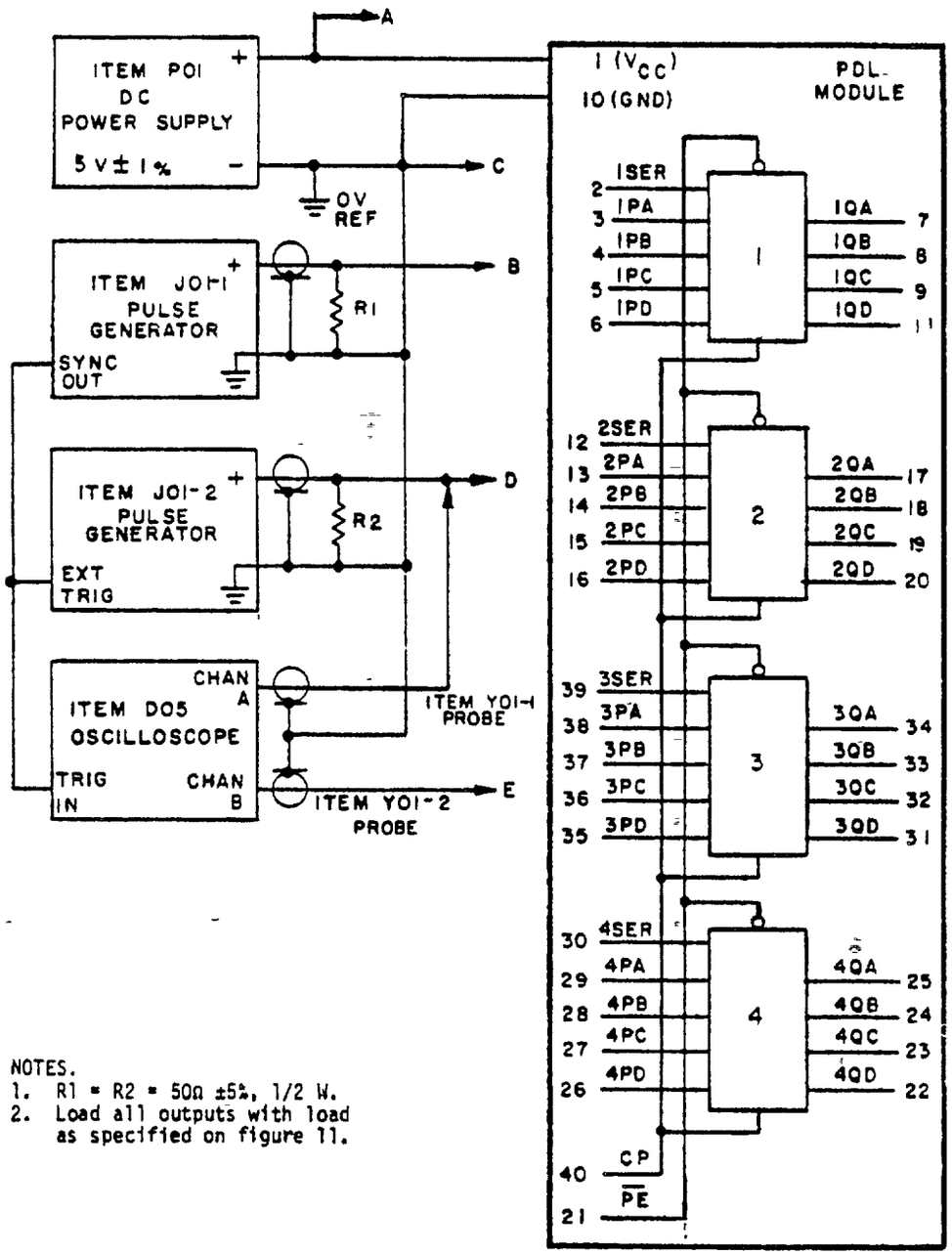
* TABLE XX. Maximum shift frequency check and minimum clock pulse width check test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	100 Maximum shift frequency check and minimum clock pulse width check	21	2	3-6	40	11	135	150	ns	15,18
T2	200 "	"	12	13-16	"	20	"	"	"	"
T3	300 "	"	39	35-38	"	31	"	"	"	"
T4	400 "	"	30	26-29	"	22	"	"	"	"

TABLE XXI. Parallel enable minimum setup and hold time check test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	Parallel enable setup time check (measure delay t_D)	21	40	7	5	45	ns	19,20
U2	"	"	"	17	"	"	"	"
U3	"	"	"	34	"	"	"	"
U4	"	"	"	25	"	"	"	"
U5	Parallel enable hold time check (measure delay t_D)	"	"	7	"	"	"	19,21
U6	"	"	"	17	"	"	"	"
U7	"	"	"	34	"	"	"	"
U8	"	"	"	25	"	"	"	"

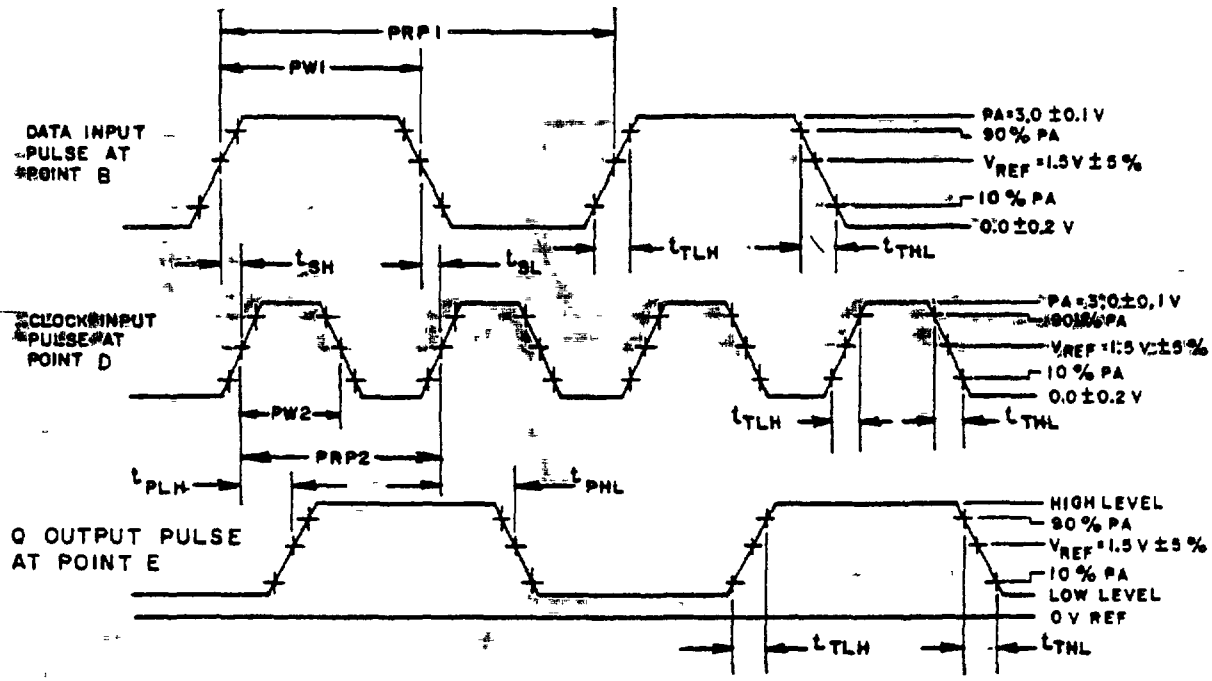
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NOTES.
1. R1 = R2 = 50Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} and t_{PLH}) and minimum pulse and sample set-up time check, sample and sample hold check, and maximum frequency test circuit and waveforms.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PW1 = 200 ±20 ns.
   PRP1 = 400 ±40 ns.

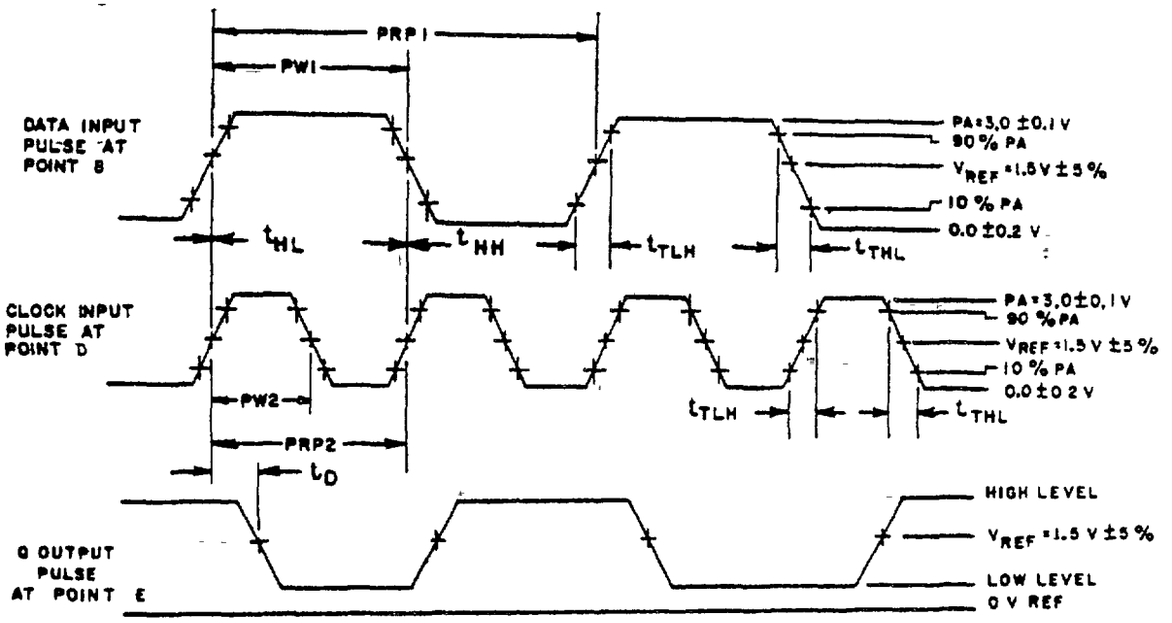
   PW2 = 100 ±10 ns.
   PRP2 = 200 ±20 ns.

   t_{SH} (setup time high) = $45\ ^{+2}_{-0}$ ns.

   t_{SL} (setup time low) = $45\ ^{+2}_{-0}$ ns.
2. Before making measurement, verify that the output high level is $\geq$ 2.4 V and the output low level is $<$ 0.4 V.
3. All voltages are referenced to ground.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) and minimum parallel and serial setup time check test circuit waveforms for tables XV through XVIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PW1 = 200 ±20 ns.
   PRP1 = 400 ±40 ns.

   PW2 = 100 ±10 ns.
   PRP2 = 200 ±20 ns.

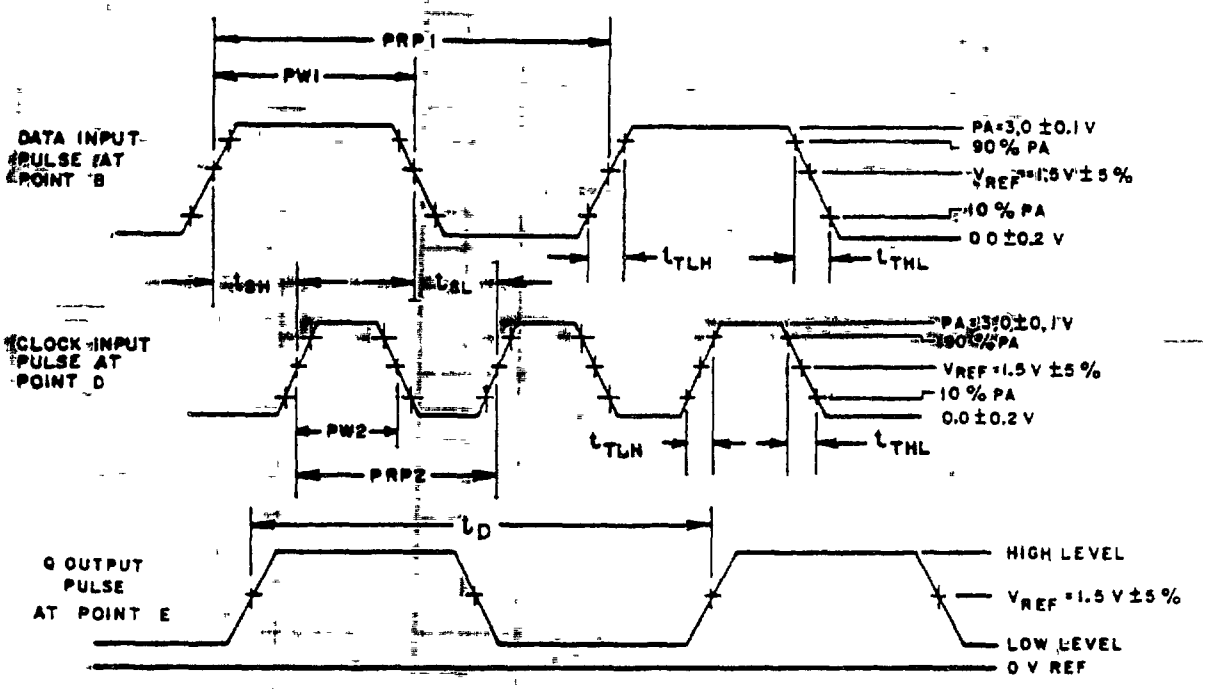
   t_{HH}(hold time high) = $0\ ^{+2}_{-0}$ ns.

   t_{HL}(hold time low) = $0\ ^{+2}_{-0}$ ns.

2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

* FIGURE 17. Parallel and serial hold time check test circuit waveforms for table XIX.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PW1 = 70 ±2 ns.
   PRP1 = 140 ±4 ns.

   PW2 = $35\ ^{+2}_{-0}$ ns.

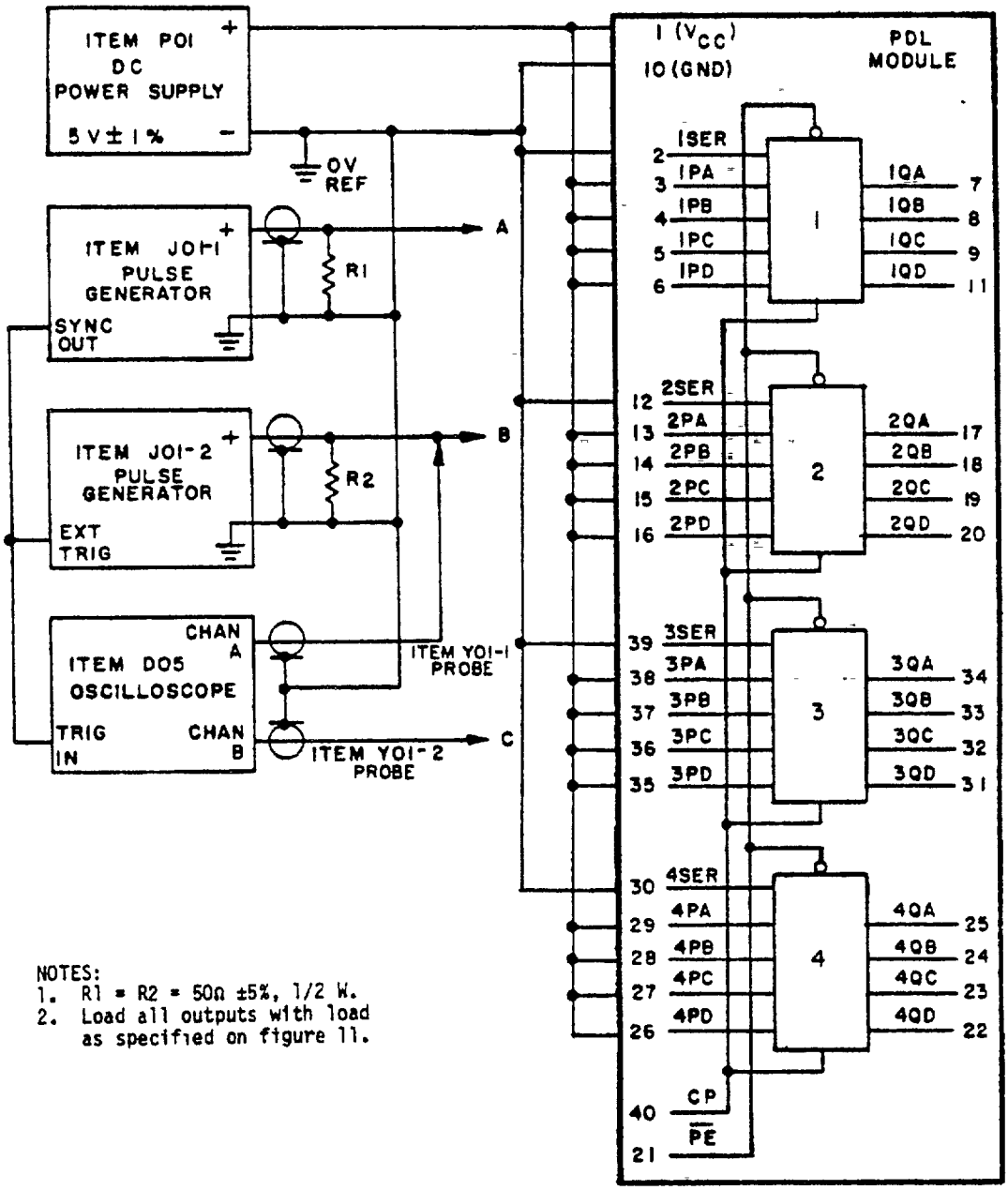
   PRP2 = $70\ ^{+4}_{-0}$ ns.

   t_{SH} (setup time high) = $45\ ^{+2}_{-0}$ ns.

   t_{SL} (setup time low) = $45\ ^{+2}_{-0}$ ns.
2. Before making measurements verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.
4. Wait a minimum of four clock pulses before making measurement.

* FIGURE 18. Maximum shift-frequency check and minimum clock pulse width check test circuit waveforms for table XX.

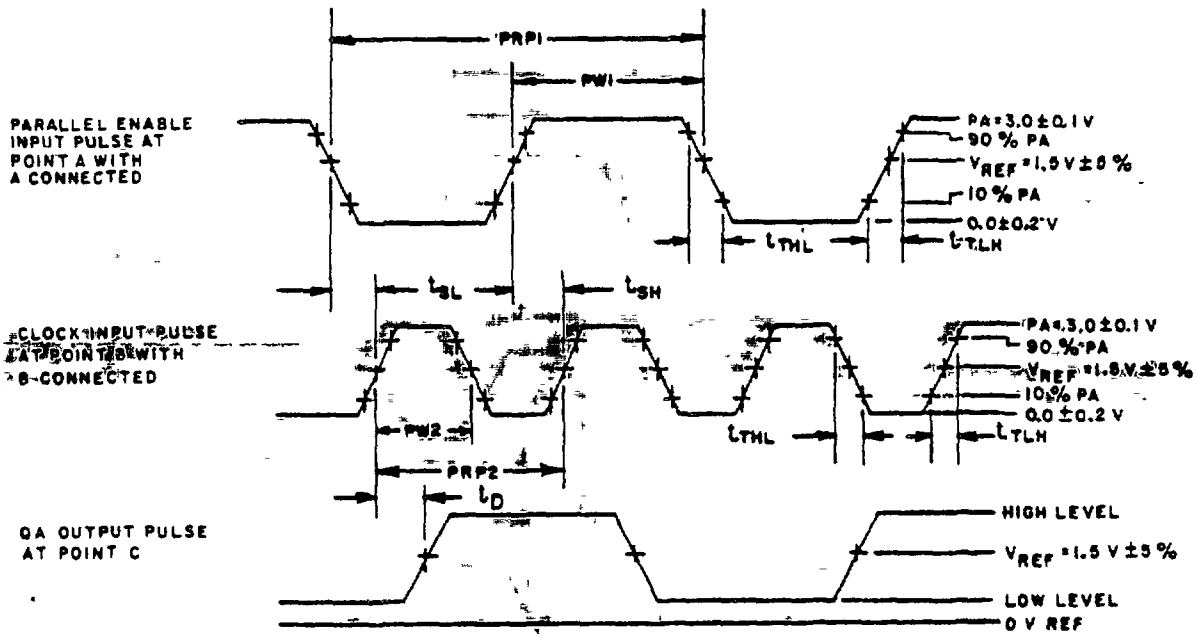
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:04Z
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

FIGURE 19. Parallel enable minimum setup and hold time check test circuit for table XXI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PW1 = 200 ±20 ns.
   PRP1 = 400 ±40 ns.

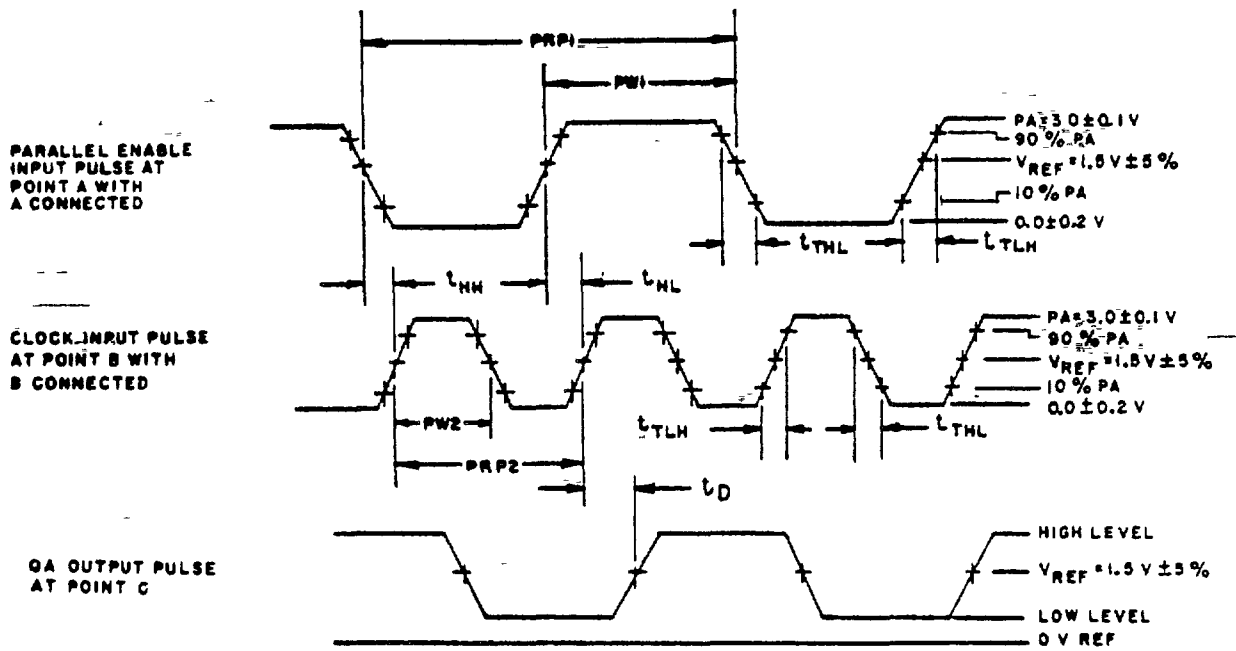
   PW2 = 100 ±10 ns.
   PRP2 = 200 ±20 ns.

   t_{SH}(setup time high) = $45\ ^{+2}_{-0}$ ns.

   t_{SL}(setup time low) = $45\ ^{+2}_{-0}$ ns.
2. Before making measurements, verify that the output high level is ≥ 2.4 V and the output low level is ≤ 0.4 V.
3. All voltages are referenced to ground.

* FIGURE 20. Parallel enable minimum setup and hold time check test circuit waveforms for table XXI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PW1 = 200 ±20 ns.
   PRP1 = 400 ±40 ns.

   PW2 = 100 ±10 ns.
   PRP2 = 200 ±20 ns.

   t_{HH} (hold time high) = $-10\ ^{+2}_{-0}$ ns.

   t_{HL} (hold time low) = $-10\ ^{+2}_{-0}$ ns.
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

* FIGURE 21. Parallel enable minimum setup and hold time check test circuit waveforms for table XXI.

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TABLE XXII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
V1	1QA Output crosstalk, low level	2,12, 30,39	3,21	4-6,13-16,26-29, 35-38	7	0.0	1.1	V	22,23
V2	1QB "	"	4,21	3,5,6,13-16,26-29,35-38	8	"	"	"	"
V3	1QC "	"	5,21	3,4,6,13-16,26-29,35-38	9	"	"	"	"
V4	1QD "	"	6,21	3-5,13-16,26-29, 35-38	11	"	"	"	"
V5	2QA "	"	13,21	3-6,14-16,26-29, 35-38	17	"	"	"	"
V6	2QB "	"	14,21	3-6,13,15,16,26-29,35-38	18	"	"	"	"
V7	2QC "	"	15,21	3-6,13,14,16,26-29,35-38	19	"	"	"	"
V8	2QD "	"	16,21	3-6,13-15,26-29, 35-38	20	"	"	"	"
V9	3QA "	"	21,38	3-6,13-16,26-29, 35-37	34	"	"	"	"
V10	3QB "	"	21,37	3-6,13-16,26-29, 35,36,38	33	"	"	"	"
V11	3QC "	"	21,36	3-6,13-16,26-29, 35,37,38	32	"	"	"	"
V12	3QD "	"	21,35	3-6,13-16,26-29, 36-38	31	"	"	"	"
V13	4QA "	"	21,29	3-6,13-16,26-28, 35-38	25	"	"	"	"

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TABLE XXII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V14	4QB Output crosstalk, low level	2,12, 30,39	21,28	3-6,13-16,26,27, 29,35-38	24	0.0	1.1	V	22,23
V15	4QC "	"	21,27	3-6,13-16,26,28, 29,35-38	23	"	"	"	"
V16	4QD "	"	21,26	3-6,13-16,27-29, 35-38	22	"	"	"	"
V17	1QA "	"	3,21	4-6,13-16,26-29, 35-38	7	0	400	mV	22,24
V18	1QB "	"	4,21	3,5,6,13-16,26-29,35-38	8	"	"	"	"
V19	1QC "	"	5,21	3,4,6,13-16,26-29,35-38	9	"	"	"	"
V20	1QD "	"	6,21	3-5,13-16,26-29, 35-38	11	"	"	"	"
V21	2QA "	"	13,21	3-6,14-16,26-29, 35-38	17	"	"	"	"
V22	2QB "	"	14,21	3-6,13,15,16,26-29,35-38	18	"	"	"	"
V23	2QC "	"	15,21	3-6,13,14,16,26-29,35-38	19	"	"	"	"
V24	2QD "	"	16,21	3-6,13-15,26-29, 35-38	20	"	"	"	"
V25	3QA "	"	21,38	3-6,13-16,26-29, 35-37	34	"	"	"	"
V26	3QB "	"	21,37	3-6,13-16,26-29, 35,36,38	33	"	"	"	"
V27	3QC "	"	21,36	3-6,13-16,26-29, 35,37,38	32	"	"	"	"

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TABLE XXII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V28	3QD Output crosstalk, low level	2,12, 30,39	21,35	3-6,13-16,26-29, 36-38	31	0	400	mV	22;24
V29	4QA "	"	21,29	2-6,13-16,26-28, 35-38	25	"	"	"	"
V30	4QB "	"	21,28	3-6,13-16,26,27, 29,35-38	24	"	"	"	"
V31	4QC "	"	21,27	3-6,13-16,26,28, 29,35-38	23	"	"	"	"
V32	4QD "	"	21,26	3-6,13-16,27-29, 35-38	22	"	"	"	"
W1	1QA Output crosstalk, high level	3	2,12, 21,39, 30	4-6,13-16,26-29, 35-38	7	"	"	"	22;23
W2	1QB "	4	"	3,5,6,13-16,26-29,35-38	8	"	"	"	"
W3	1QC "	5	"	3,4,6,13-16,26-29,35-38	9	"	"	"	"
W4	1QD "	6	"	3-5,13-16,26-29,35-38	11	"	"	"	"
W5	2QA "	13	"	3-6,14-16,26-29, 35-38	17	"	"	"	"
W6	2QB "	14	"	3-6,13,15,16,26-29,35-38	18	"	"	"	"
W7	2QC "	15	"	3-6,13,14,16,26-29,35-38	19	"	"	"	"
W8	2QD "	16	"	3-6,13-15,26-29, 35-38	20	"	"	"	"

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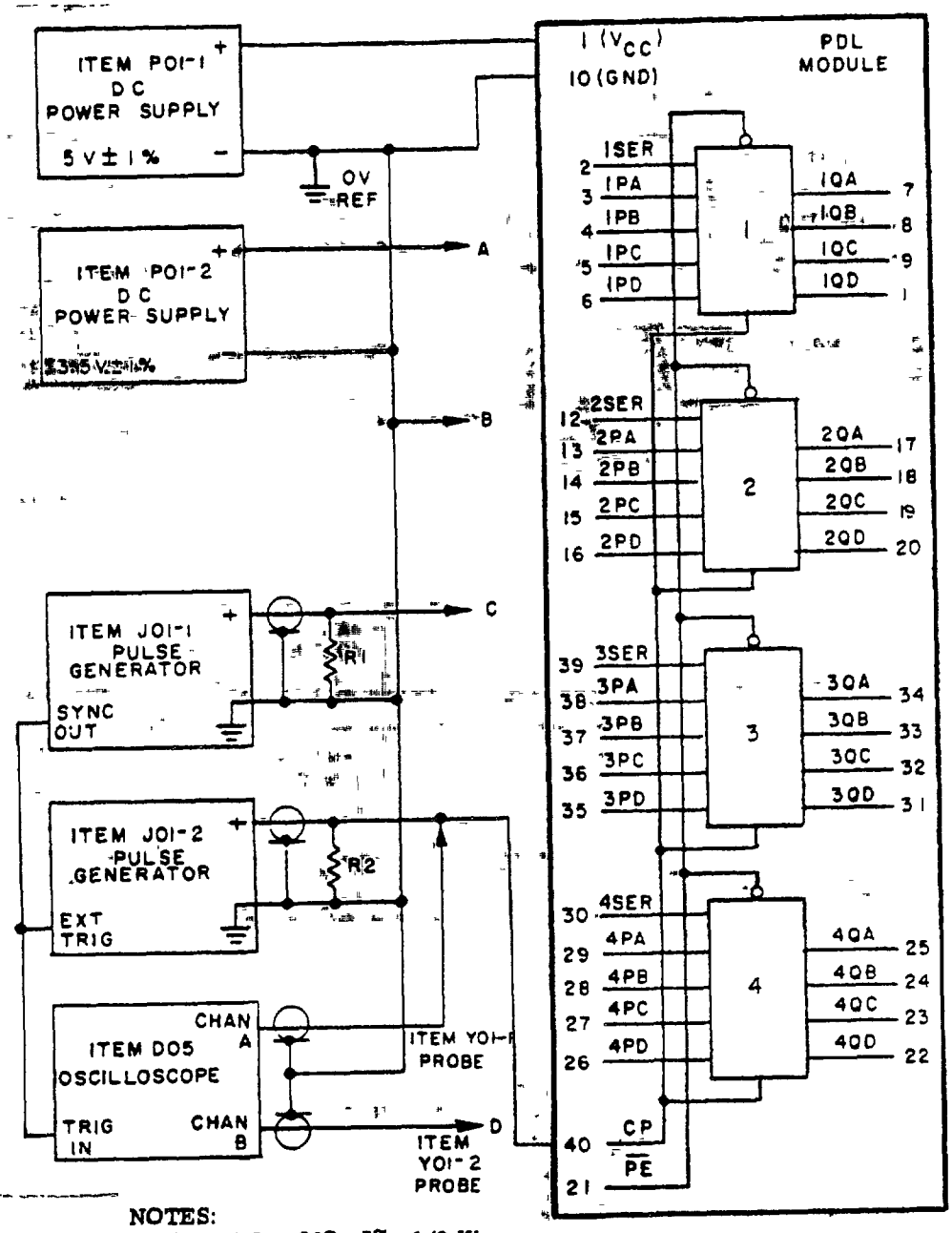
TABLE XXII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W9	3QA Output crosstalk, high level	38	2,12, 21,39, 30	3-6,13-16,26-29, 35-37	34	0	400	mV	22,23
W10	3QB "	37	"	3-6,13-16,26-29, 35,36,38	33	"	"	"	"
W11	3QC "	36	"	3-6,13-16,26-29, 35,37,38	32	"	"	"	"
W12	3QD "	35	"	3-6,13-16,26-29, 36-38	31	"	"	"	"
W13	4QA "	29	"	3-6,13-16,26-28, 35-38	25	"	"	"	"
W14	4QB "	28	"	3-6,13-16,26,27, 29,35-38	24	"	"	"	"
W15	4QC "	27	"	3-6,13-16,26,28, 29,35-38	23	"	"	"	"
W16	4QD "	26	"	3-6,13-16,27-29, 35-38	22	"	"	"	"

TABLE XXIII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
X	Power supply transient amplitude	1	0	400	mV	25,26

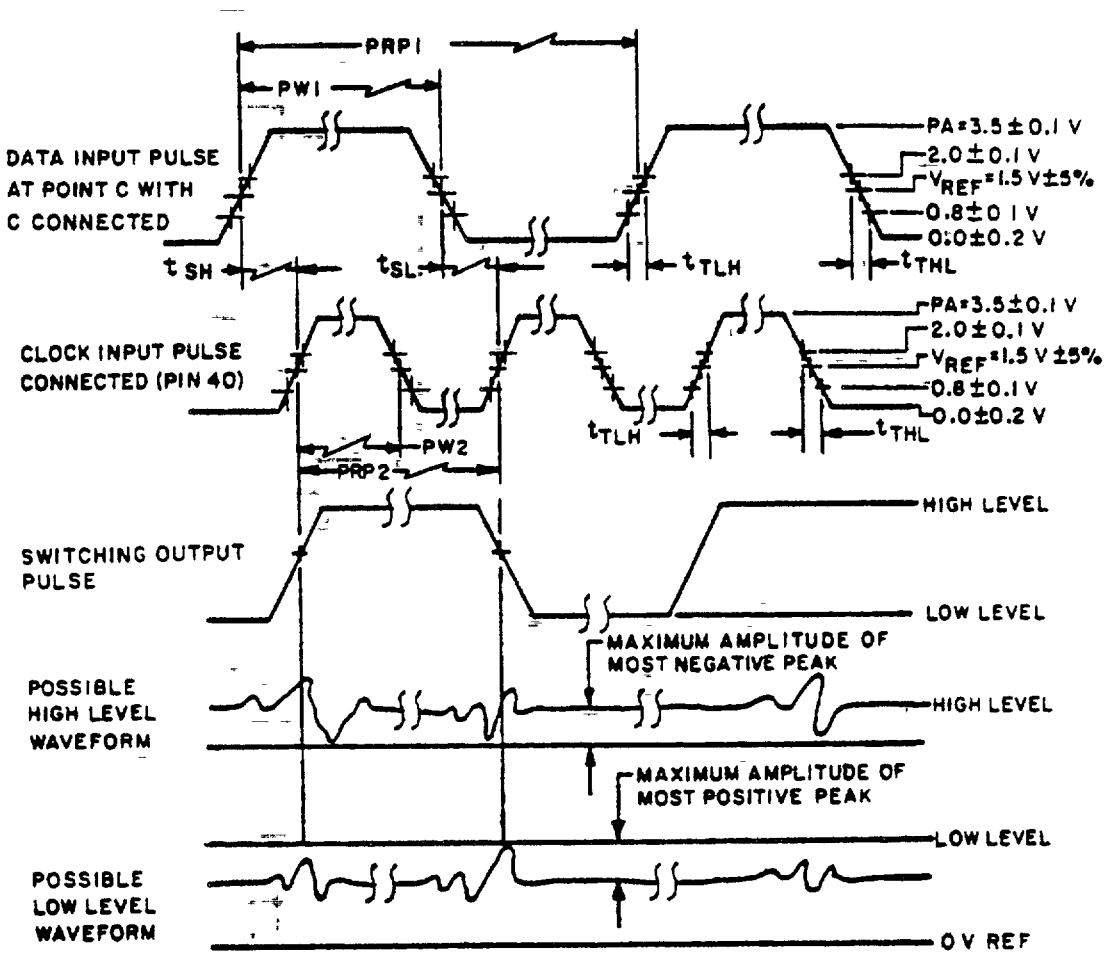
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

FIGURE 22. Output crosstalk (low-level and high-level) test circuit for table XXII.

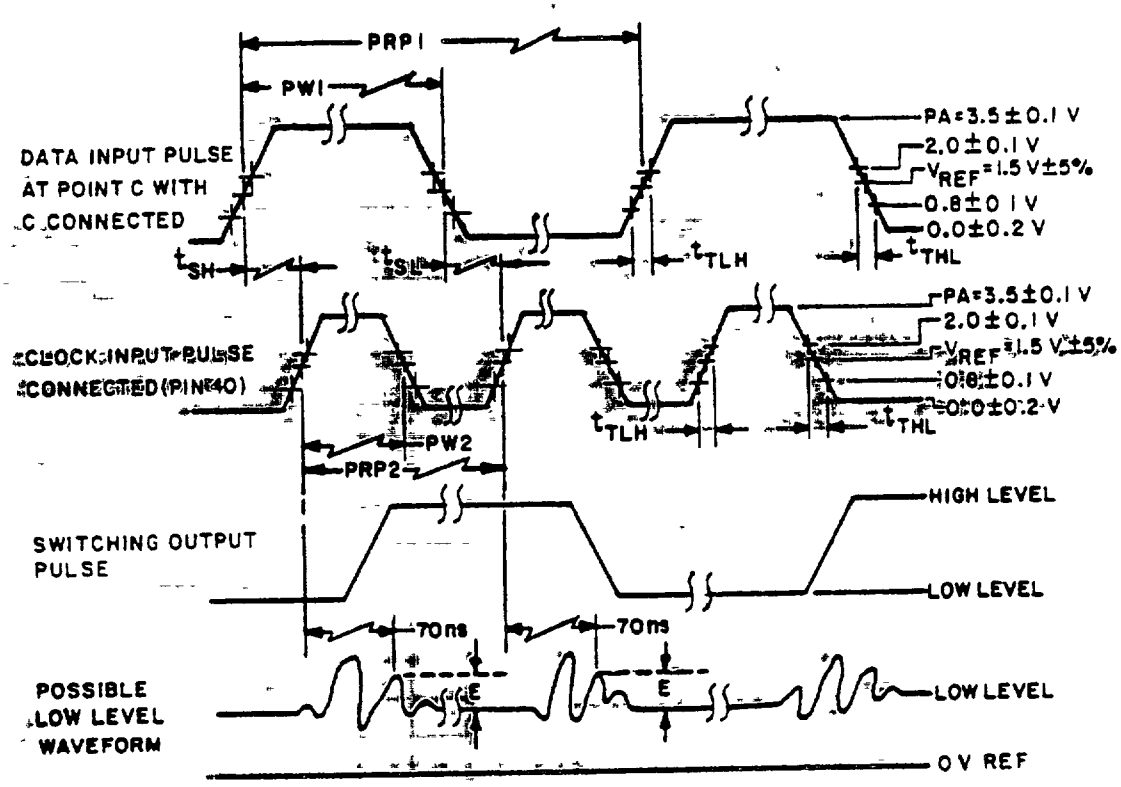
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PW1 = 500 ±50 ns.
PRP1 = 1,000 ±100 ns.
$t_{SH} = t_{SL} = 50 \pm 5$ ns.
PW2 = 100 ±10 ns.
PRP2 = 500 ±50 ns.

* FIGURE 23. Output crosstalk (low-level and high-level) test circuit waveforms for table XXII.

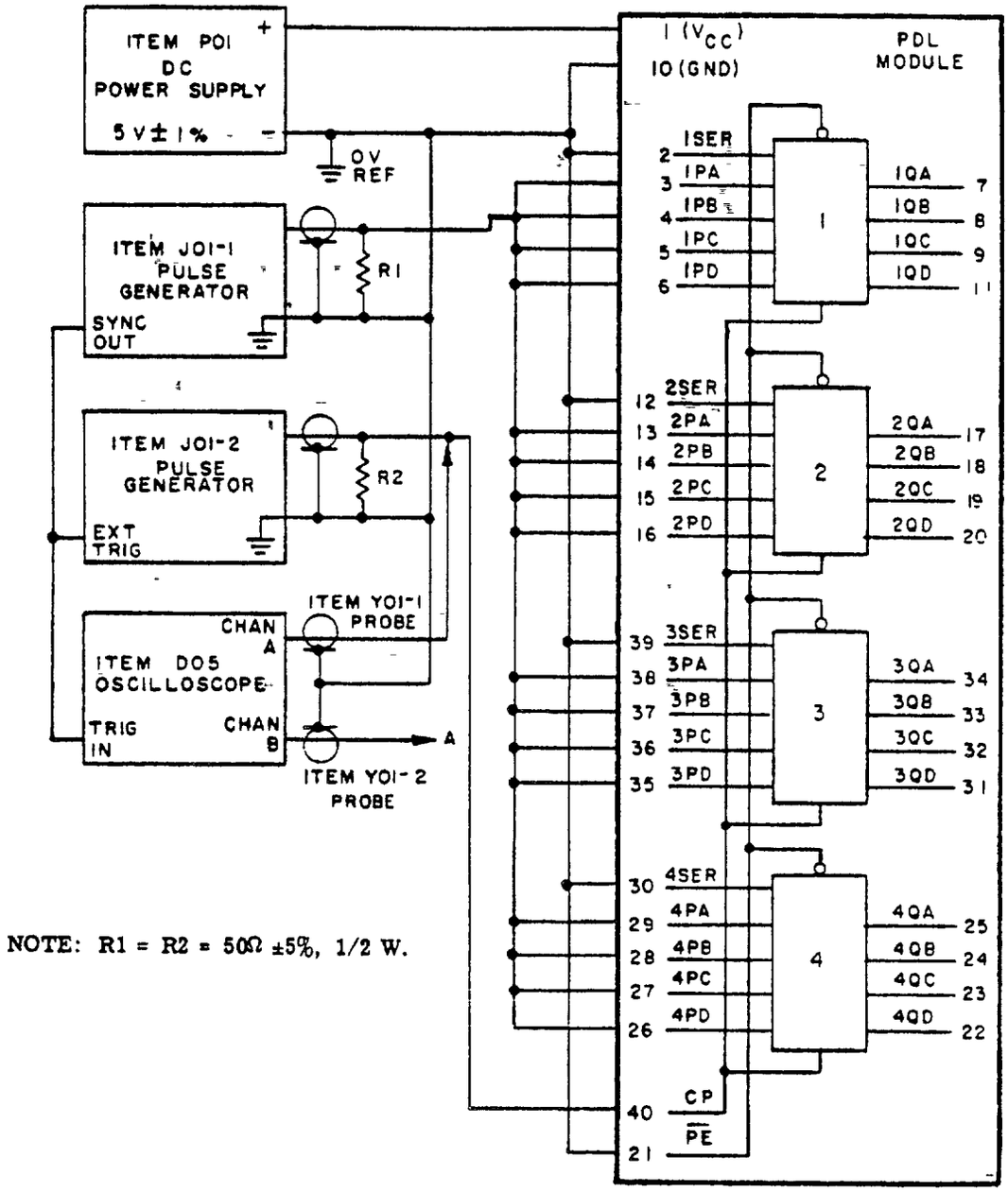
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PW1 = 500 ±50 ns.
   PRP1 = 1,000 ±100 ns.
   PW2 = 100 ±10 ns.
   PRP2 = 500 ±50 ns.
   $t_{SH} = t_{SL} = 50 \pm 50$ ns.
2. E = Maximum amplitude of most positive peak which occurs a minimum of 70 ns from the positive edge of the clock pulse.

* FIGURE 24. Output crosstalk (low-level) test circuit waveforms for tables XXII.

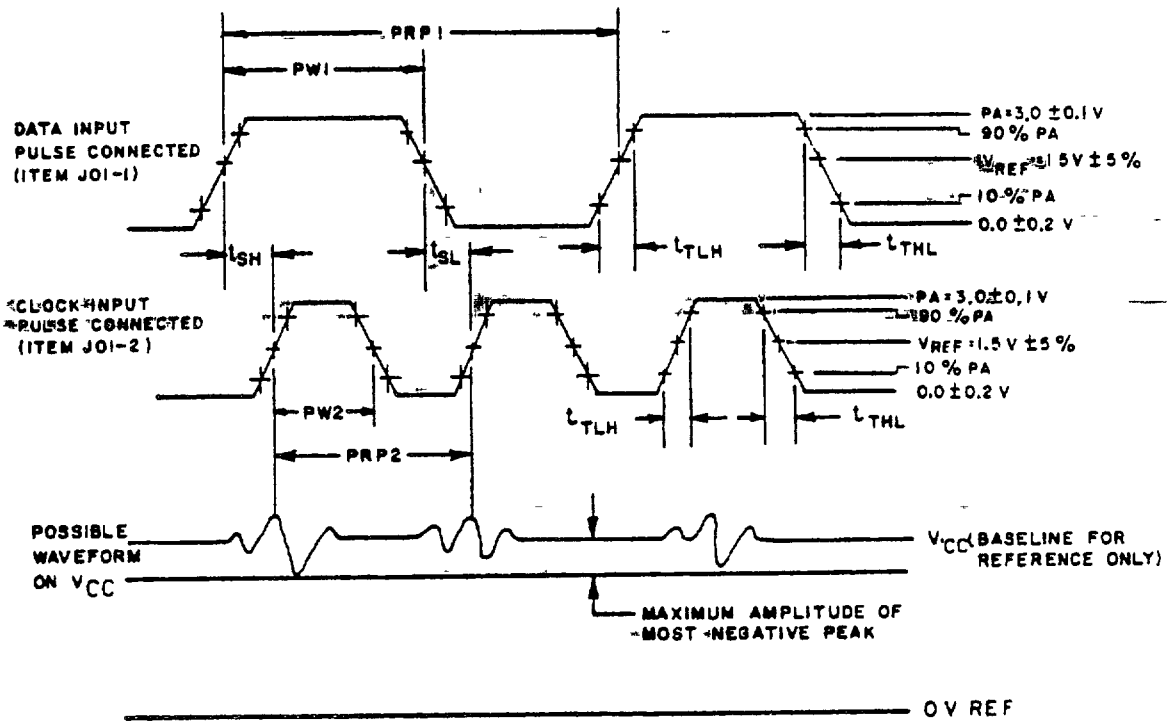
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NOTE: R1 = R2 = 50Ω ±5%, 1/2 W.

FIGURE 25. Power supply transient amplitude test circuit for table XXIII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.
PW1 = 500 ±50 ns.
PRP1 = 1,000 ±100 ns.
PW2 = 250 ±25 ns.
PRP2 = 500 ±50 ns
t_{SH} (setup time high) = 100 ±10 ns.
t_{SL} (setup time low) = 100 ±10 ns.

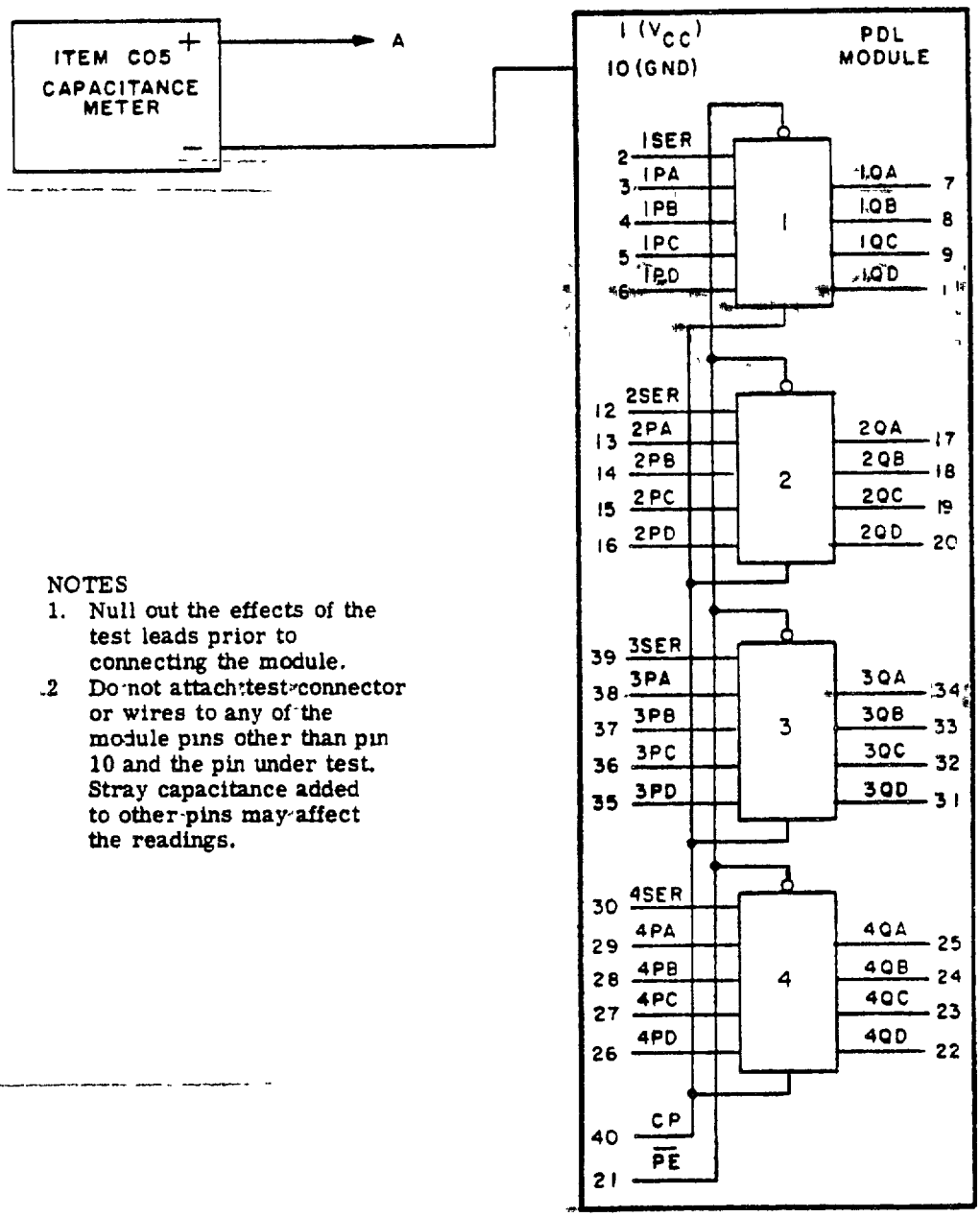
* FIGURE 26. Power supply transient amplitude test circuit waveforms for table XXIII.

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TABLE XXIV. Input capacitance test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
Y1	1SER	Input capacitance	2	2	30	pF	27
Y2	2SER	"	12	"	"	"	"
Y3	3SER	"	39	"	"	"	"
Y4	4SER	"	30	"	"	"	"
Y5	1PA	"	3	"	15	"	"
Y6	1PB	"	4	"	"	"	"
Y7	1PC	"	5	"	"	"	"
Y8	1PD	"	6	"	"	"	"
Y9	2PA	"	13	"	"	"	"
Y10	2PB	"	14	"	"	"	"
Y11	2PC	"	15	"	"	"	"
Y12	2PD	"	16	"	"	"	"
Y13	3PA	"	38	"	"	"	"
Y14	3PB	"	37	"	"	"	"
Y15	3PC	"	36	"	"	"	"
Y16	3PD	"	35	"	"	"	"
Y17	4PA	"	29	"	"	"	"
Y18	4PB	"	28	"	"	"	"
Y19	4PC	"	27	"	"	"	"
Y20	4PD	"	26	"	"	"	"
Y21	$\overline{PE}$	"	21	"	60	"	"
Y22	CP	"	40	"	60	"	"

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NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 27. Input capacitance test circuit for table XXIV.

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